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(54) **POSITION ESTIMATION SELECTION
SCHEME FOR POSITION ESTIMATION OF
A USER EQUIPMENT**

(52) **U.S. Cl.**
CPC *H04W 64/00* (2013.01); *H04L 5/0051*
(2013.01); *H04W 24/10* (2013.01)

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(57) **ABSTRACT**

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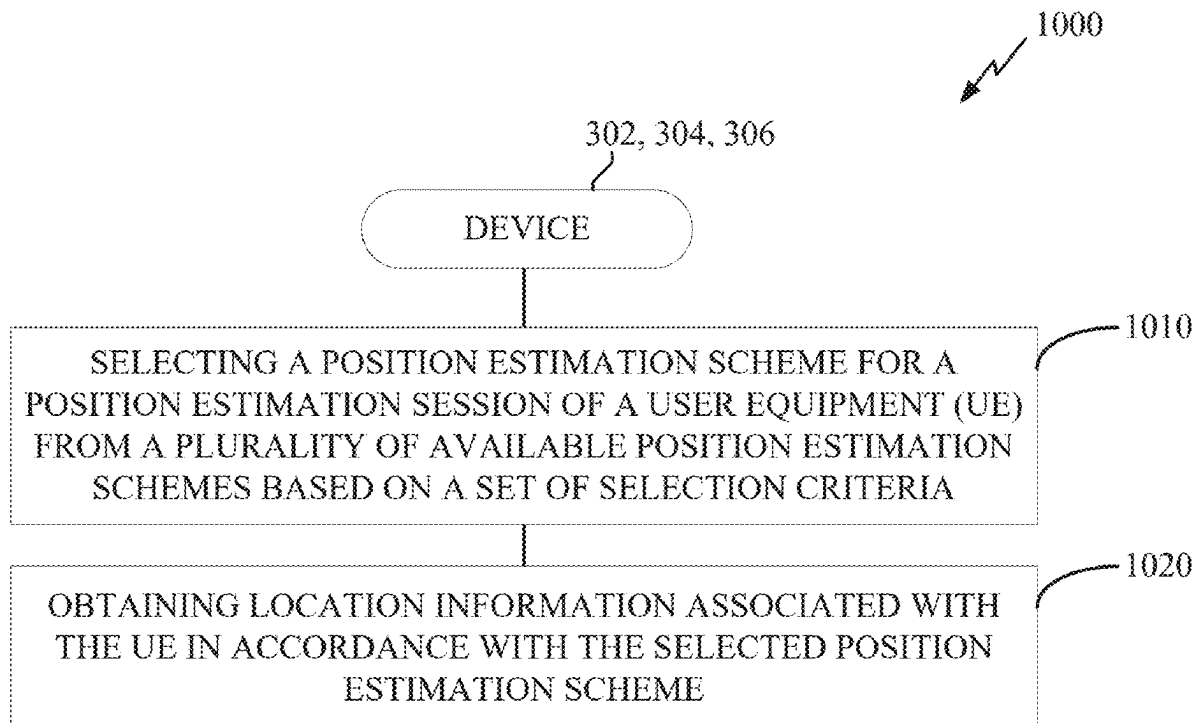
Aspects of the disclosure are directed to a position estimation selection scheme for position estimation of a user equipment (UE). For example, depending on various factors (e.g., UE battery level, positioning accuracy requirements, etc.), different positioning schemes (e.g., positioning reference signal (PRS)-only or legacy positioning, passive UE position estimation, semi-passive UE position estimation, etc.) may be preferred. In an aspect, a set of selection criteria may be evaluated to select the particular position estimation scheme to be implemented for a particular position estimation session of a UE. Such aspects may provide various technical advantages, such as reducing power consumption at the target UE (e.g., particularly advantageous for RedCap UEs), more accurate UE position estimation, reduced UE position estimation latency, and so on.

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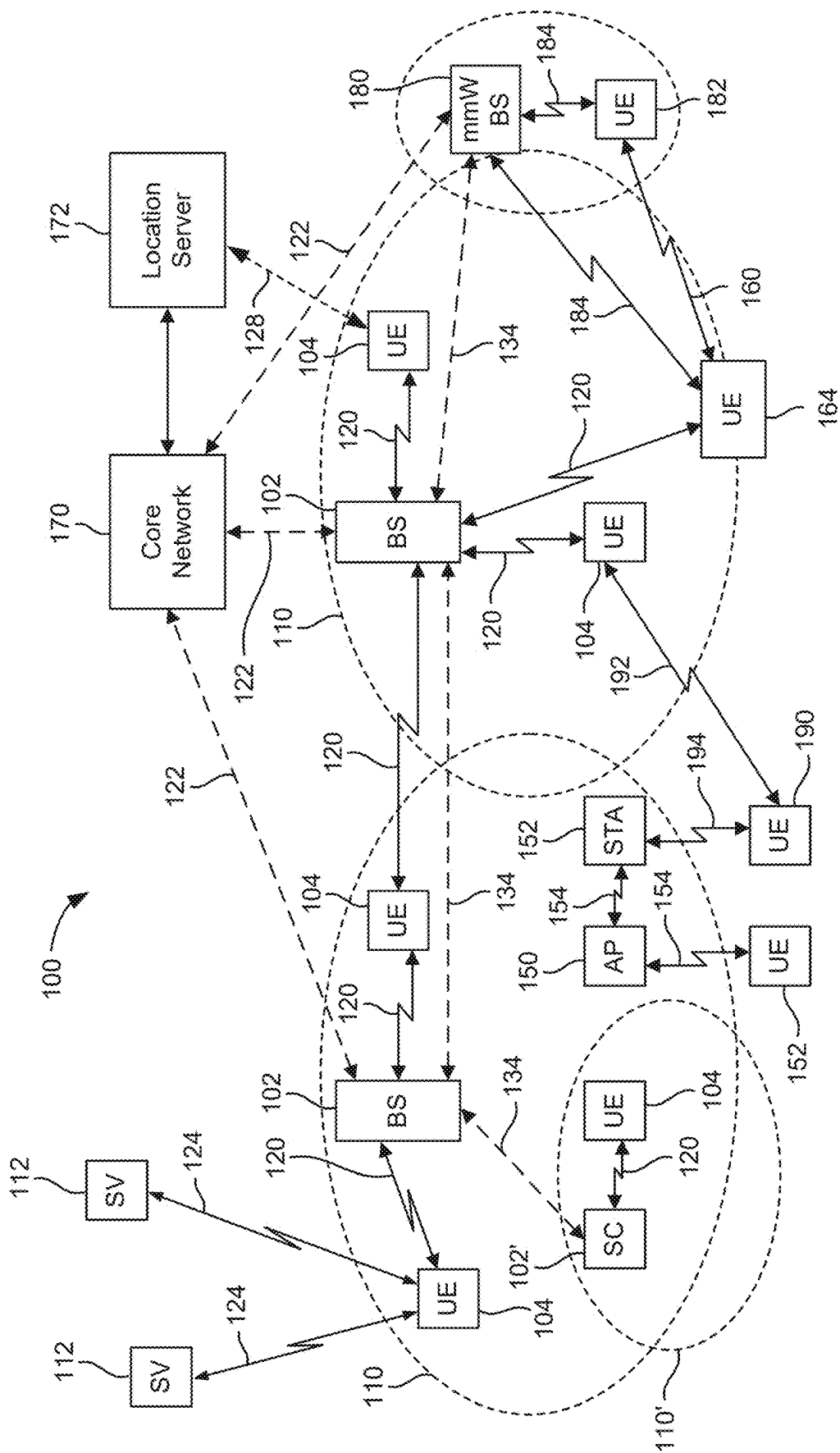


FIG. 1

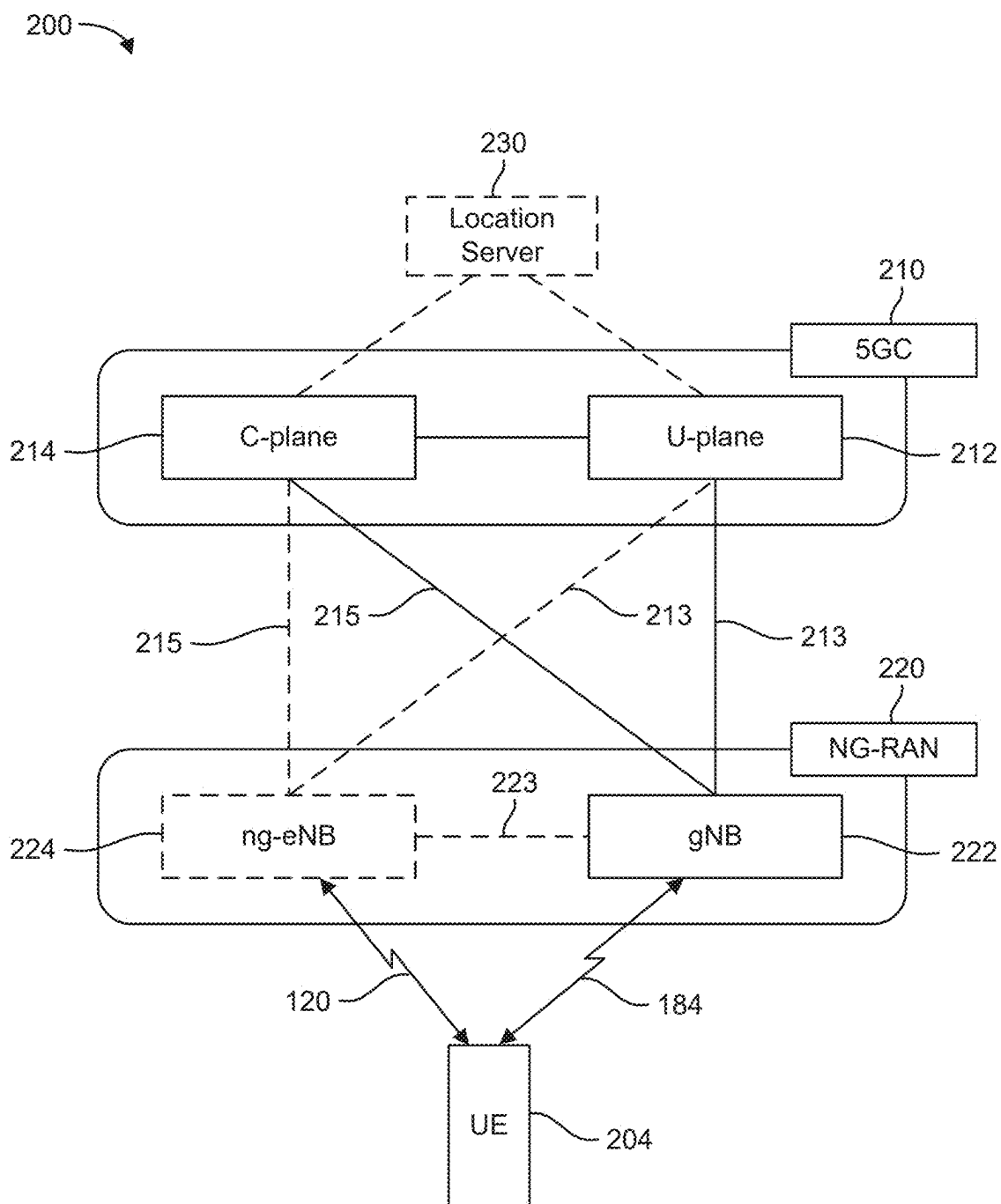


FIG. 2A

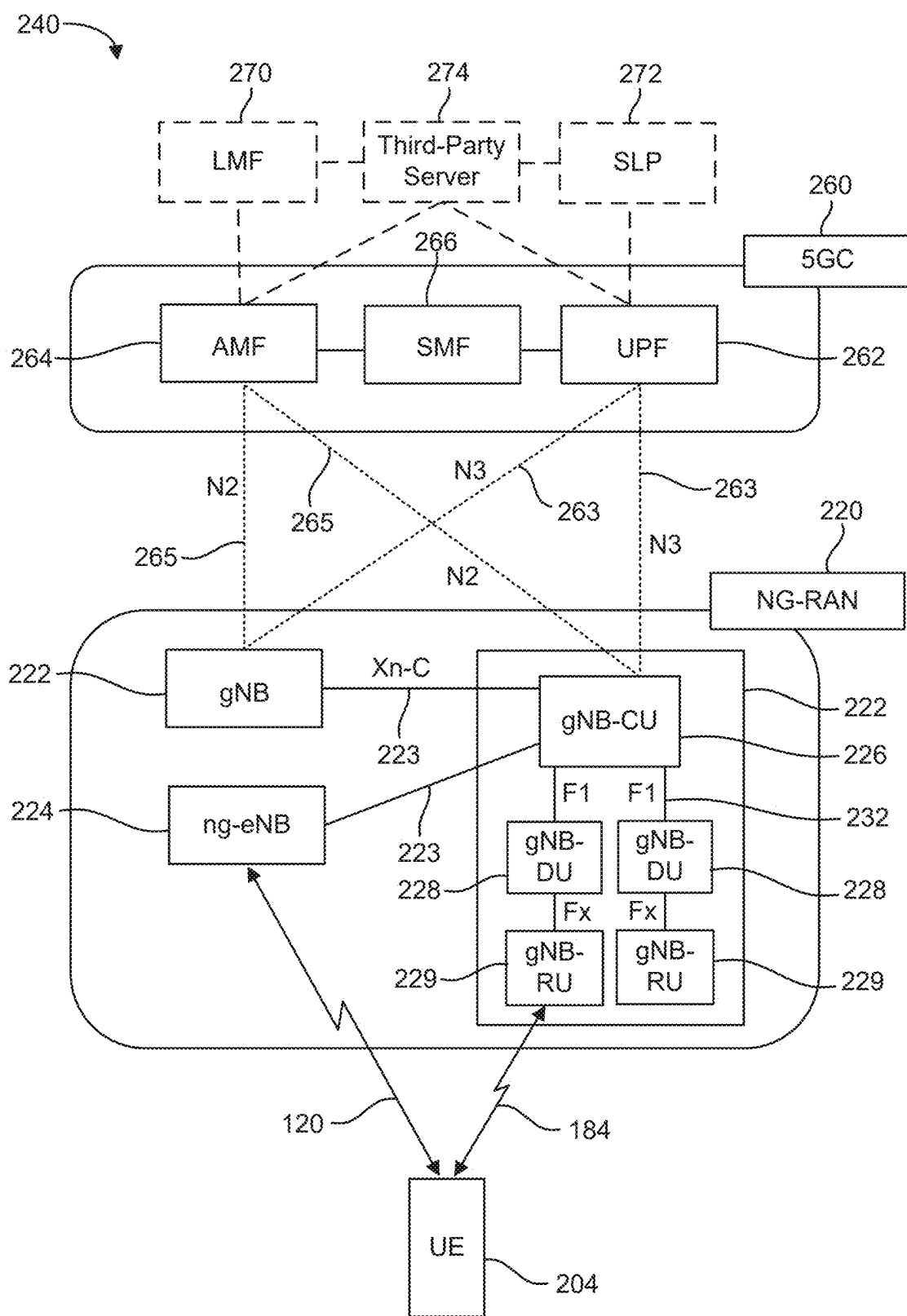
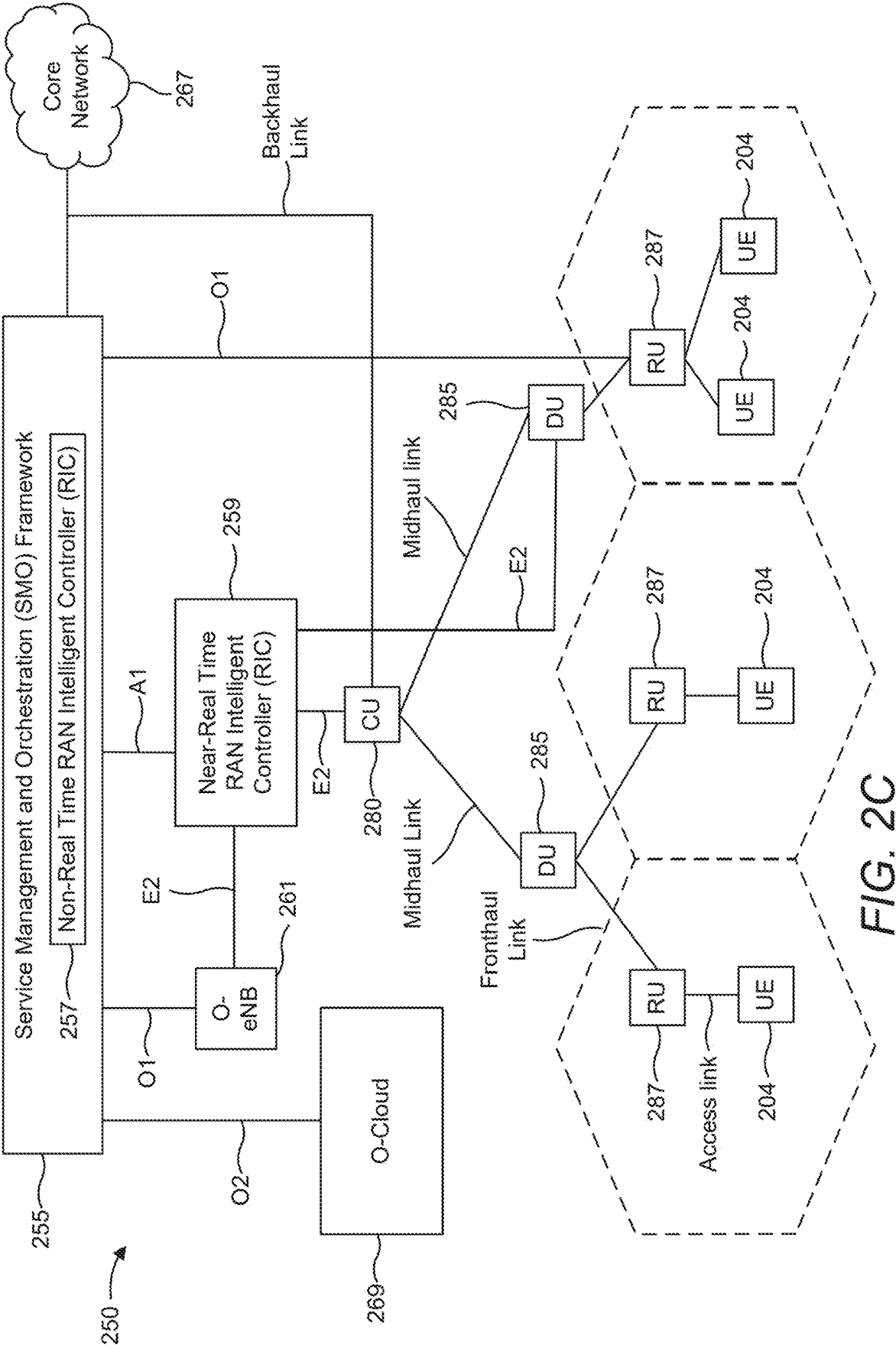


FIG. 2B



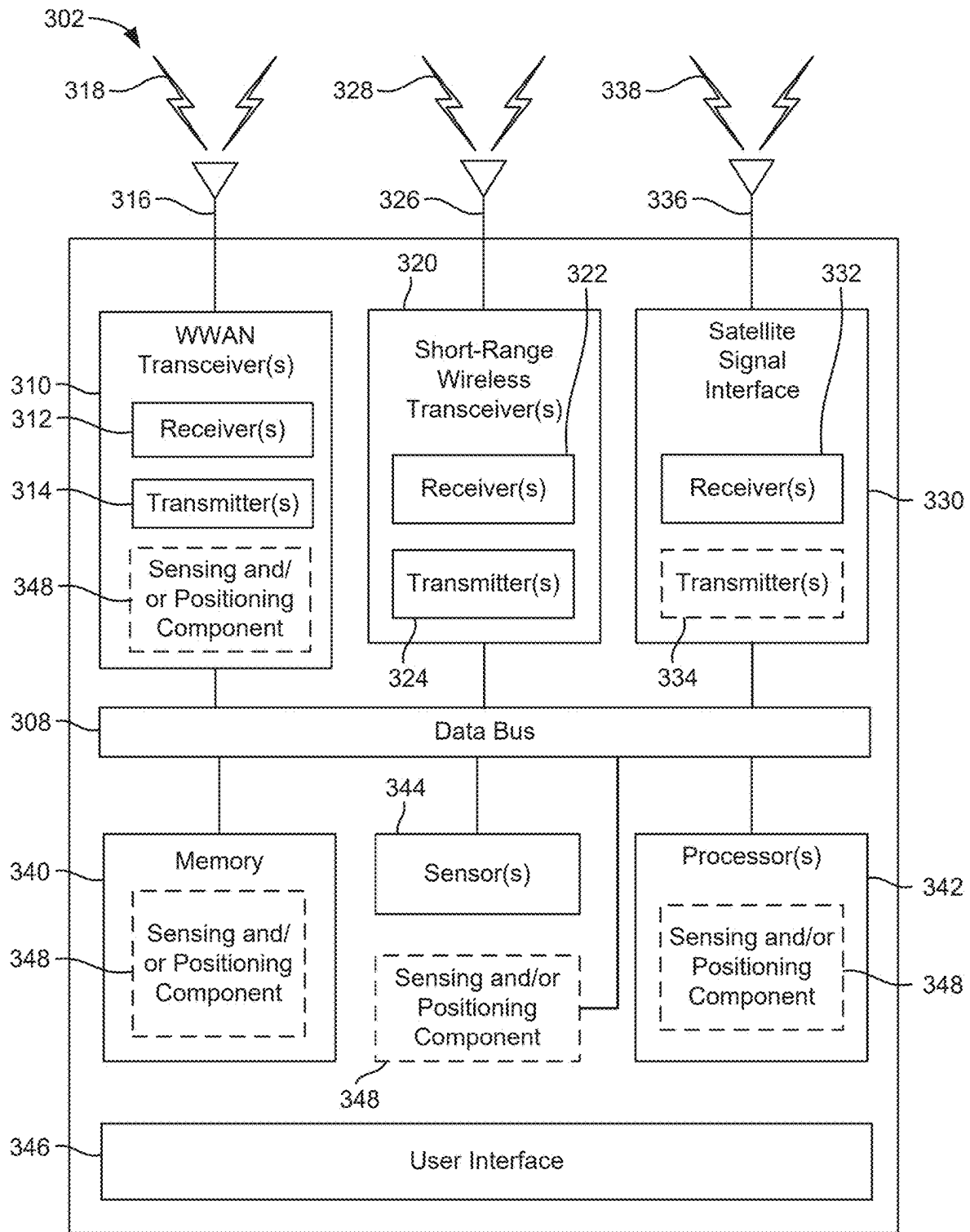


FIG. 3A

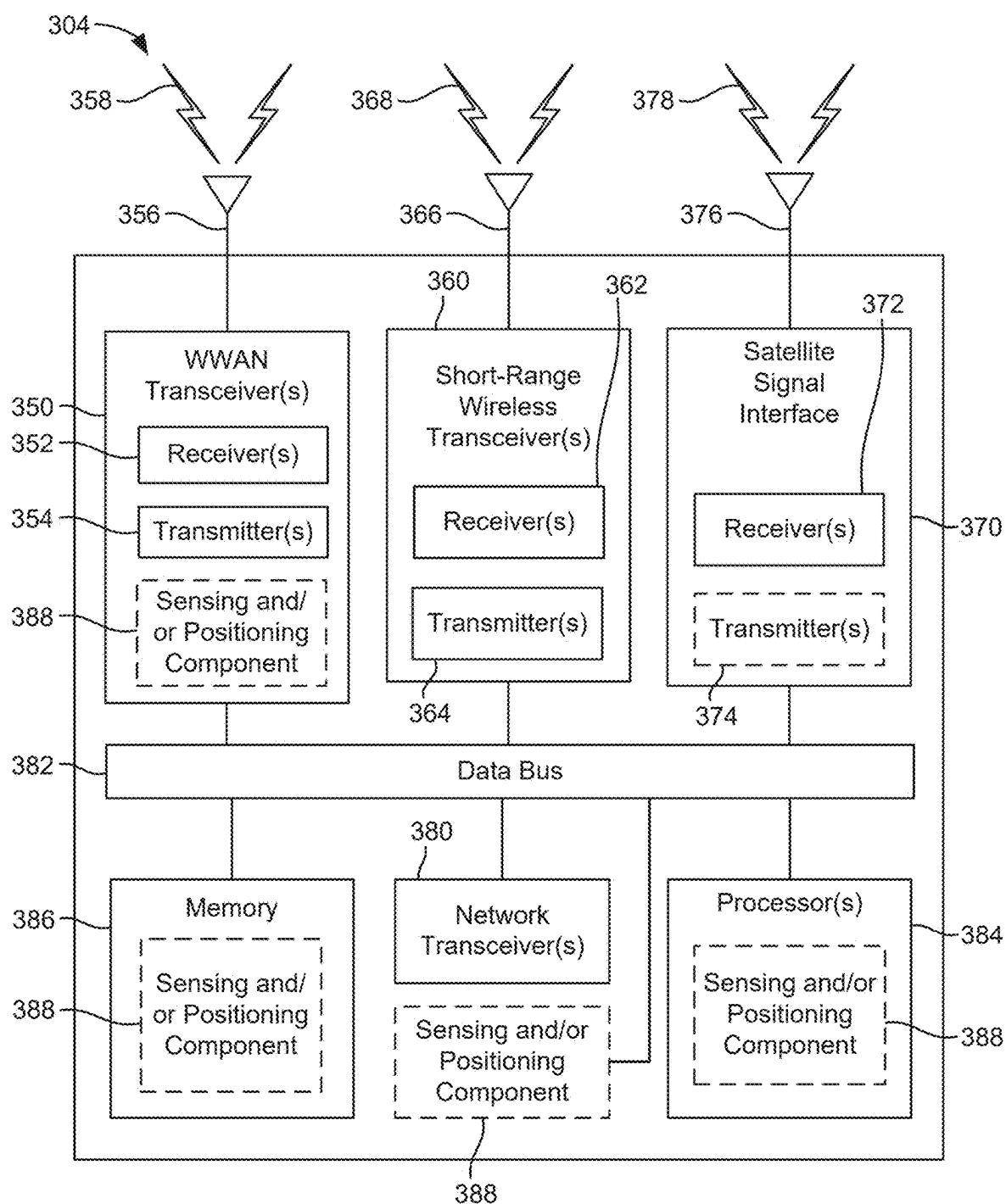
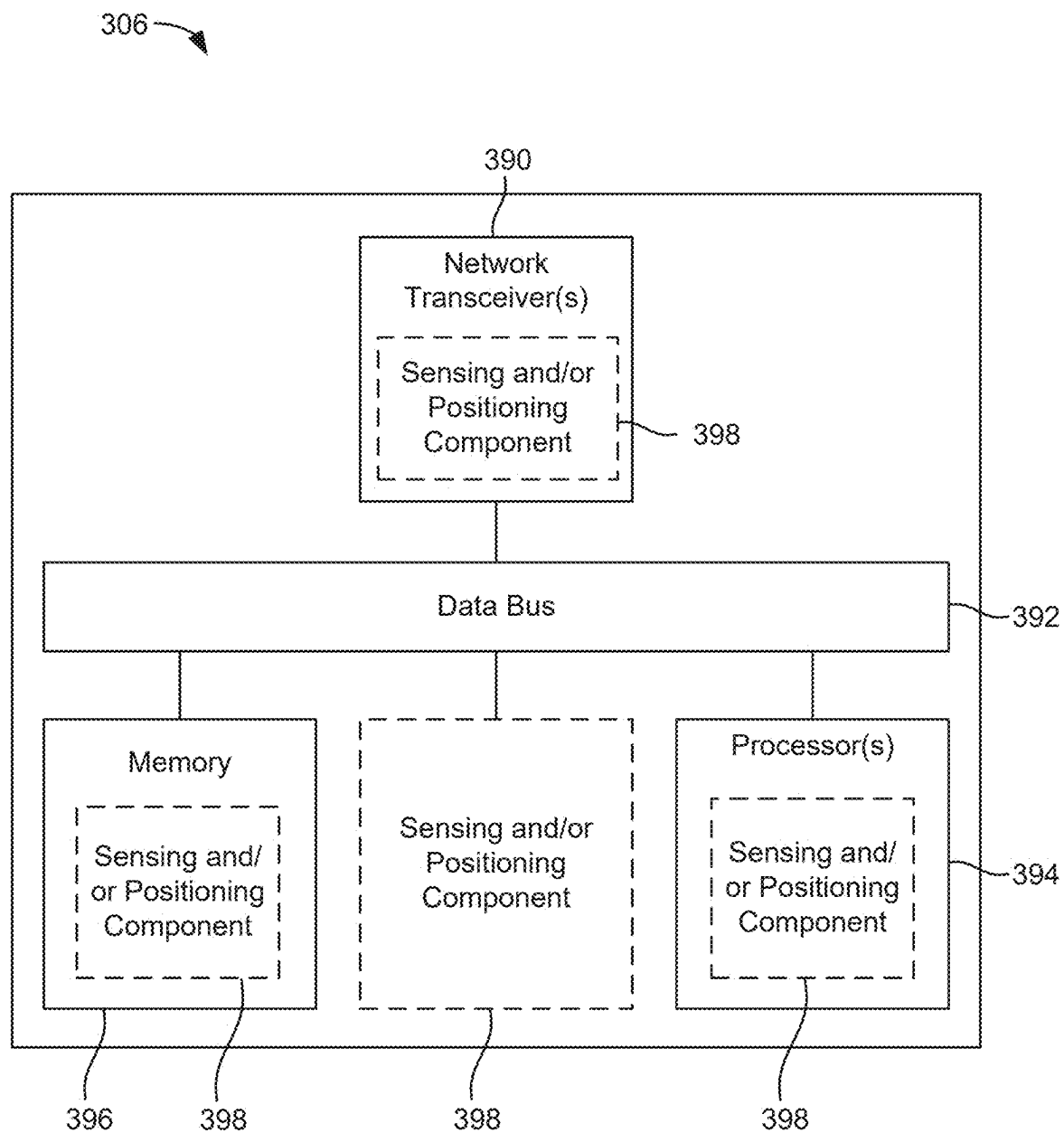
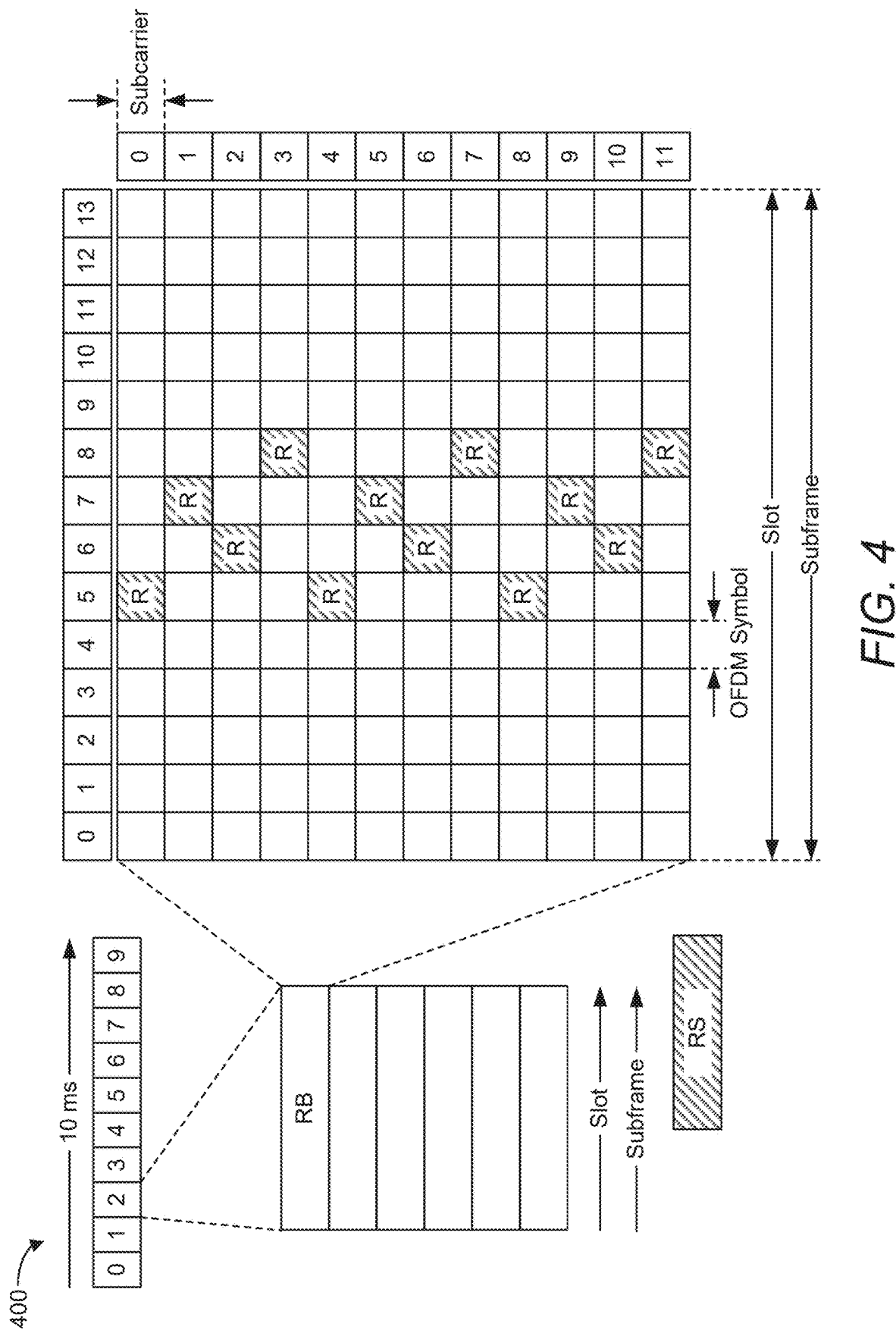


FIG. 3B

*FIG. 3C*



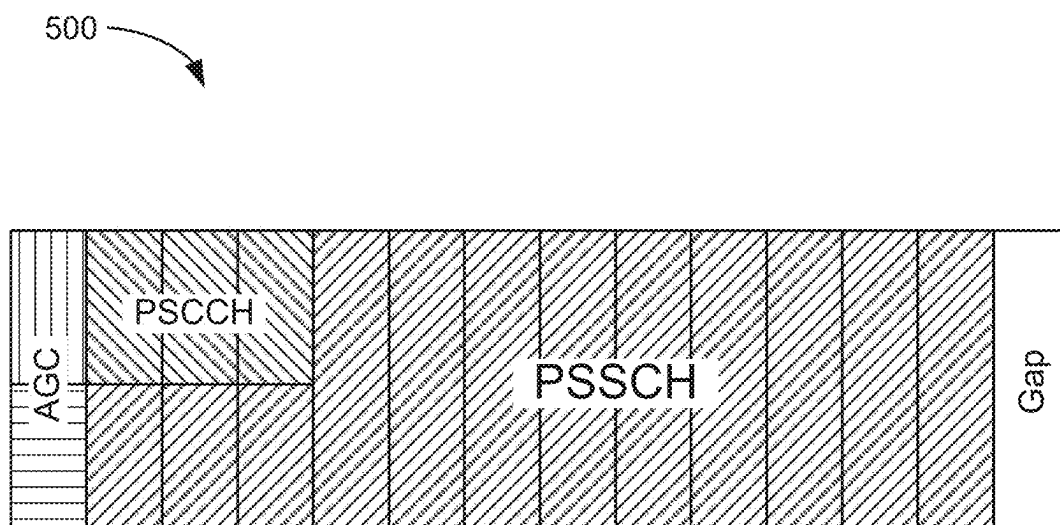


FIG. 5A

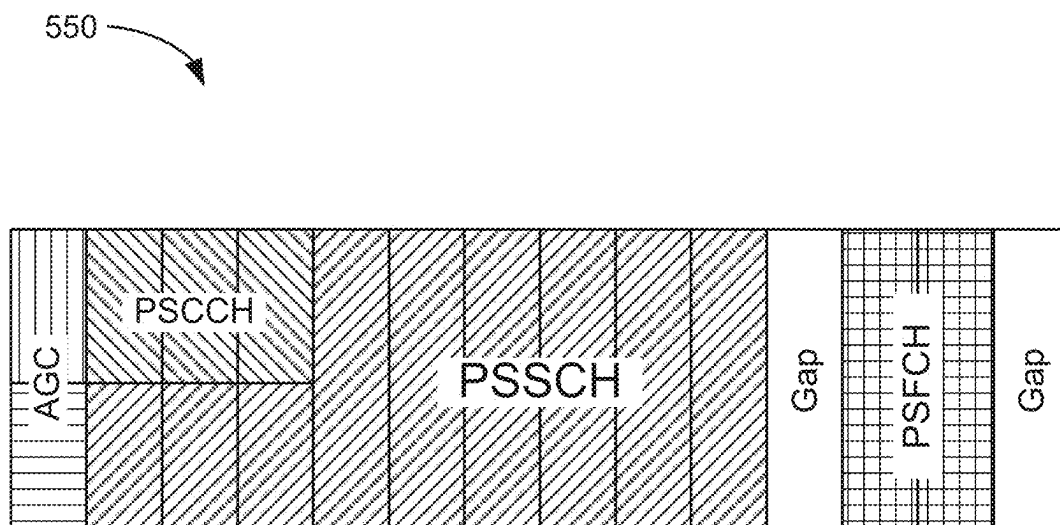


FIG. 5B

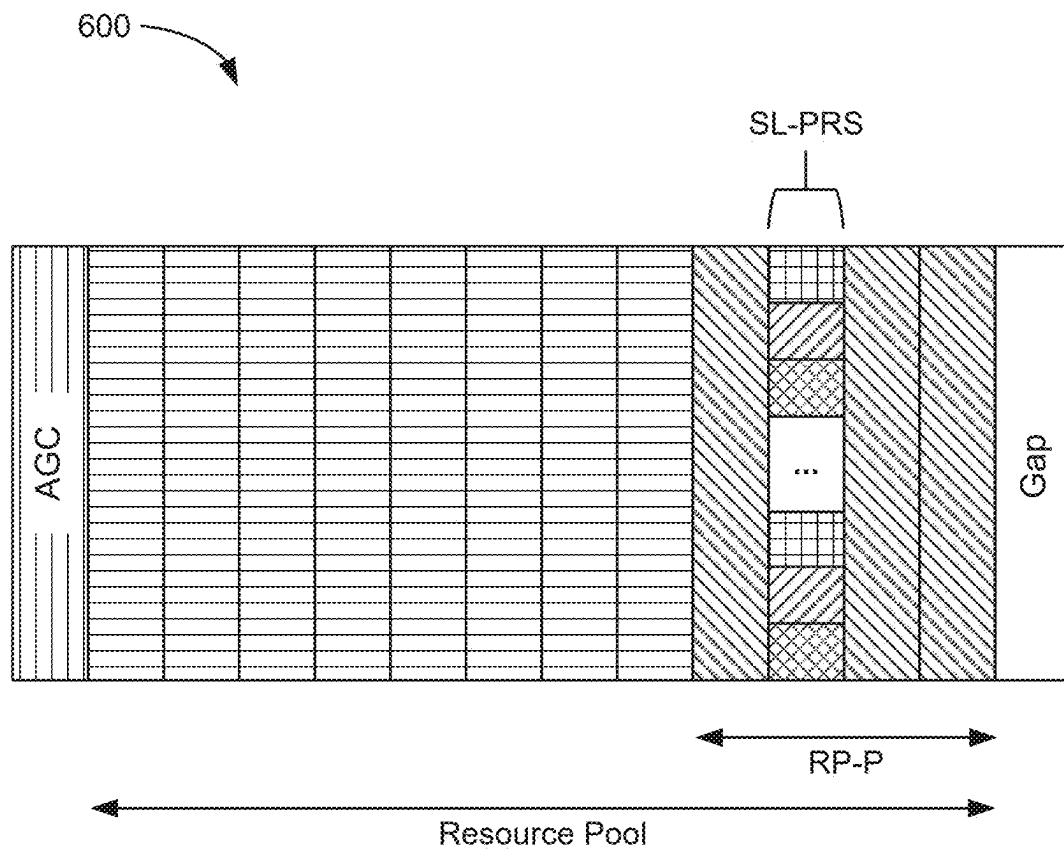


FIG. 6A

630

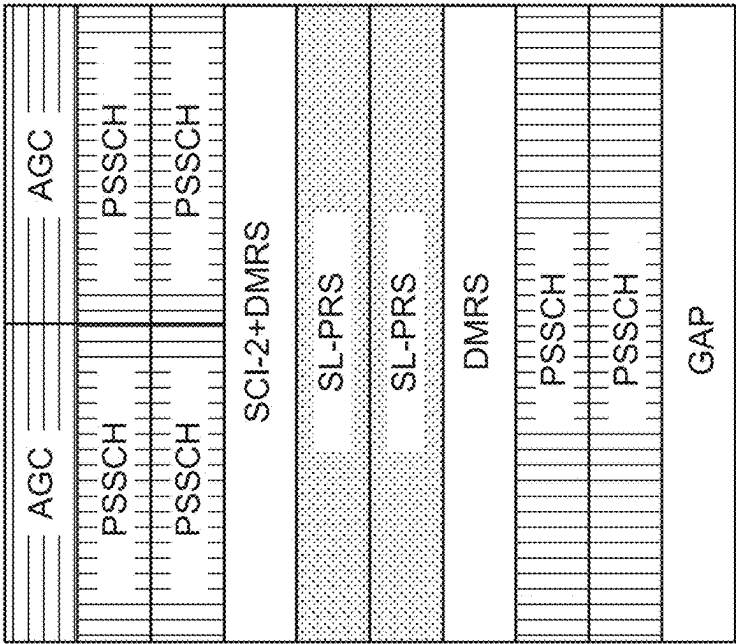


FIG. 6B

650

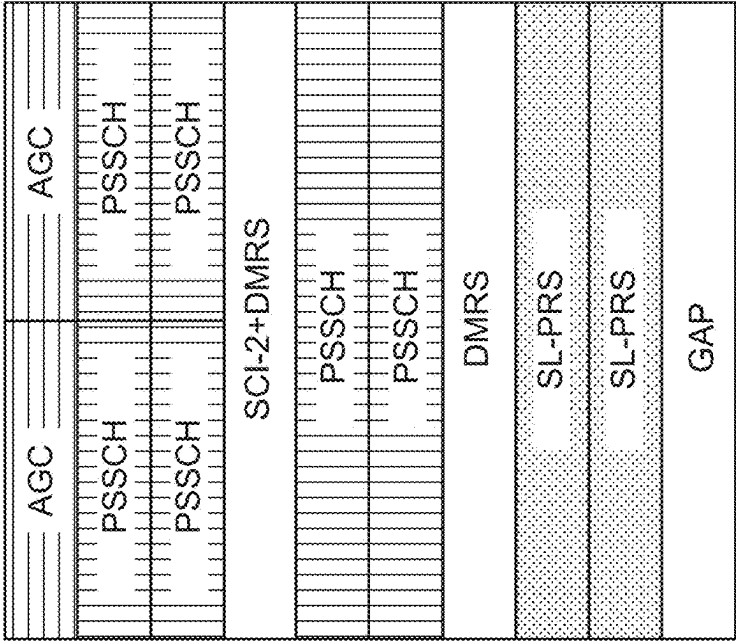


FIG. 6C

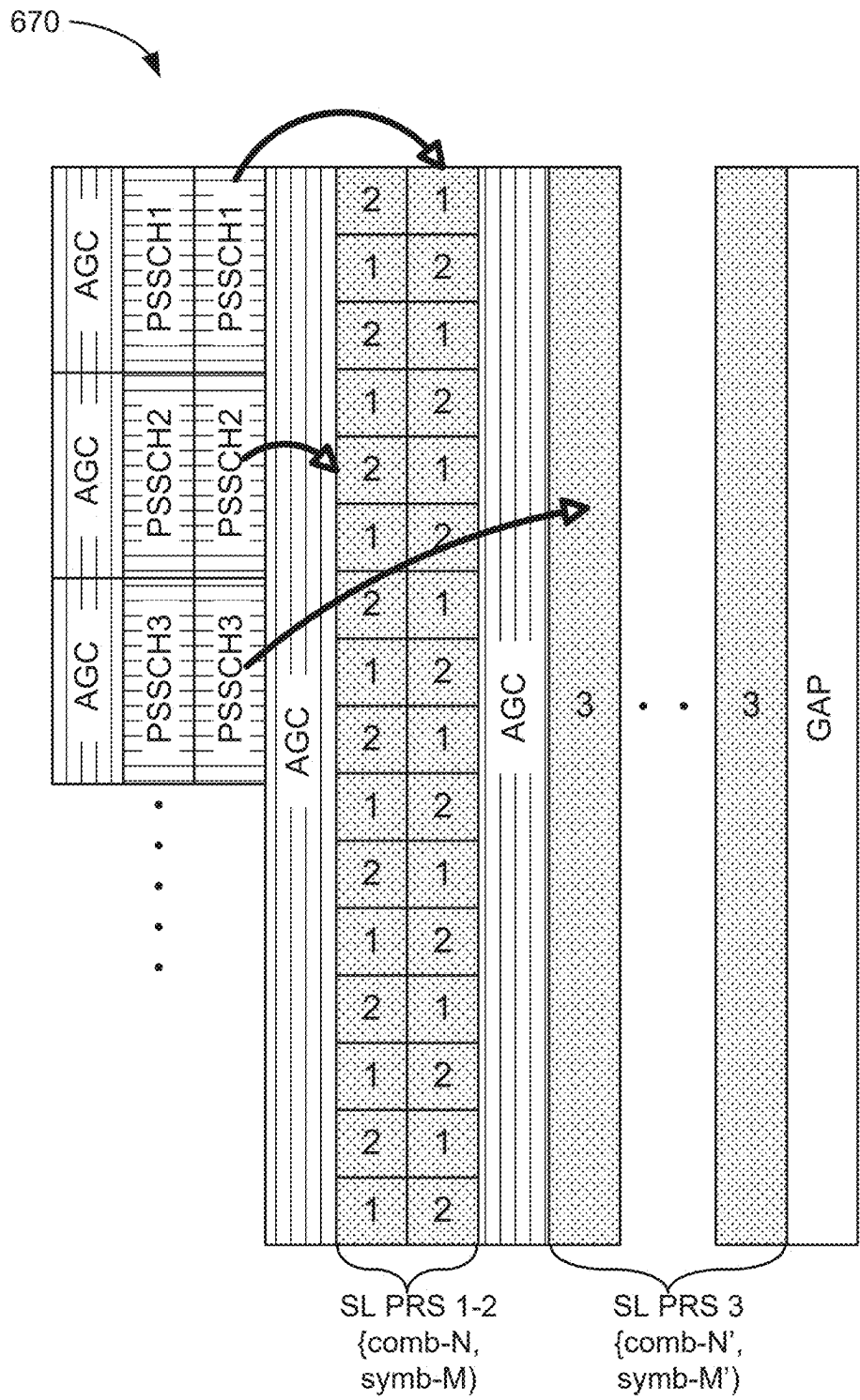


FIG. 6D

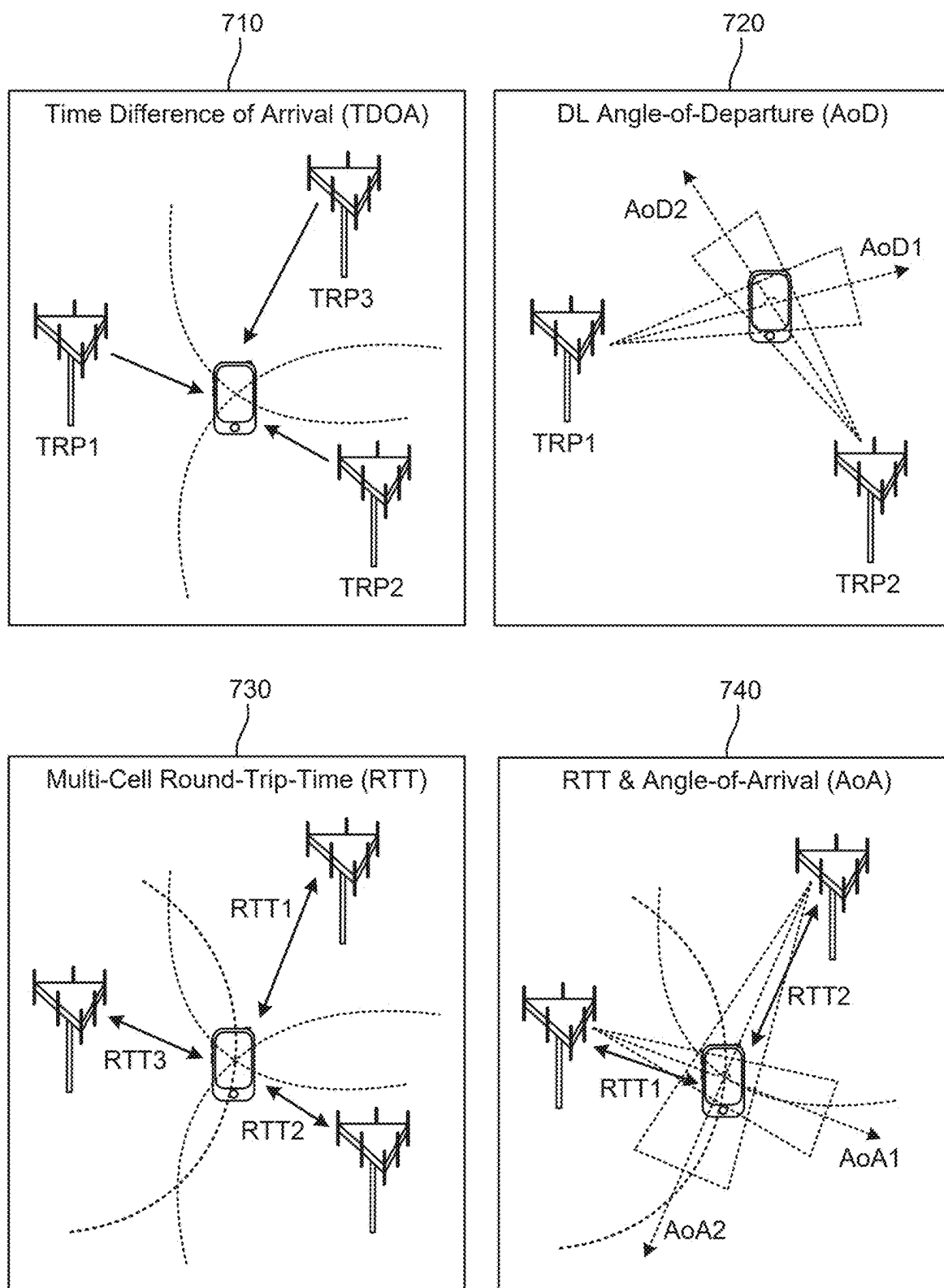


FIG. 7

800

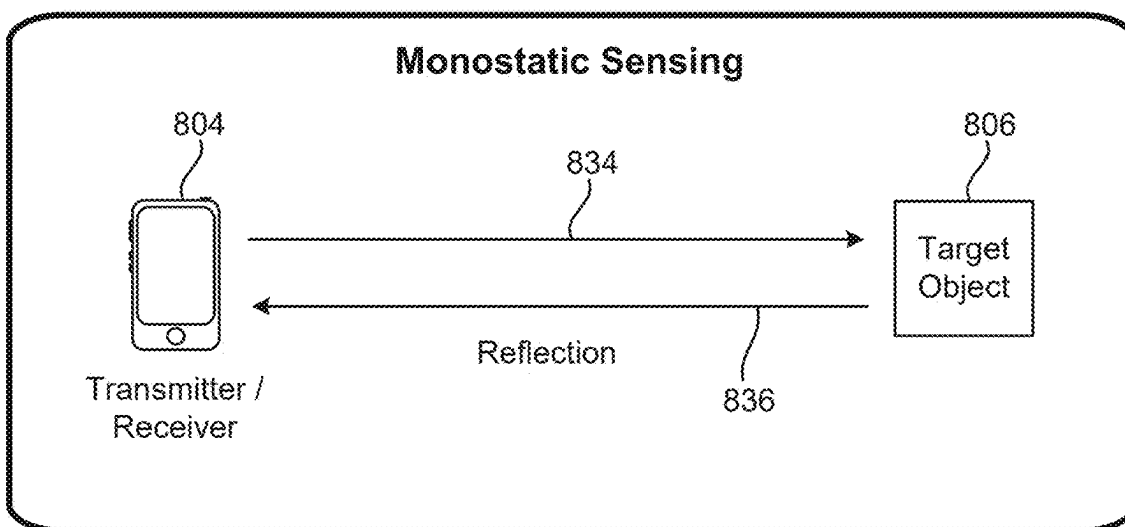


FIG. 8A

830

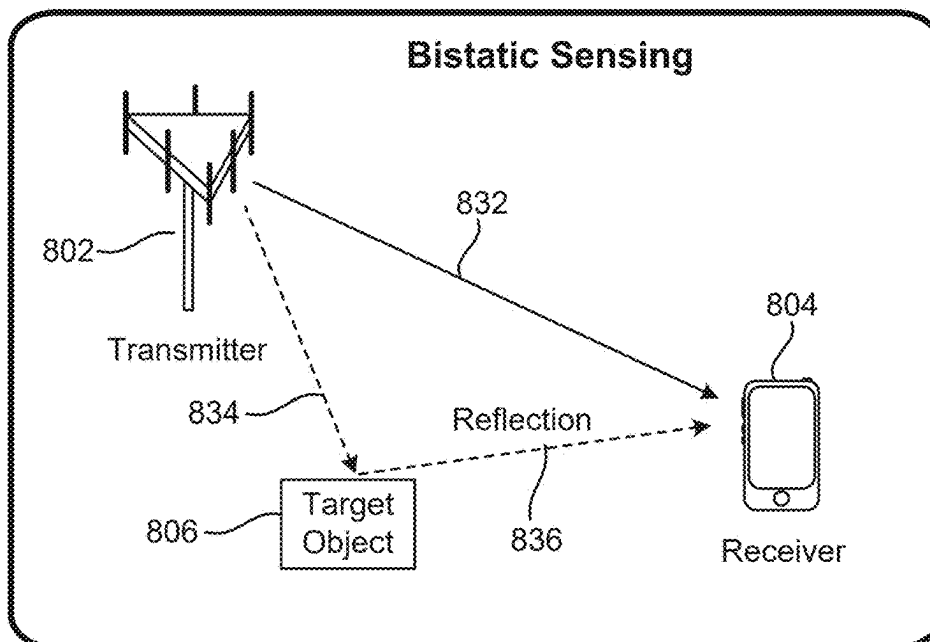


FIG. 8B

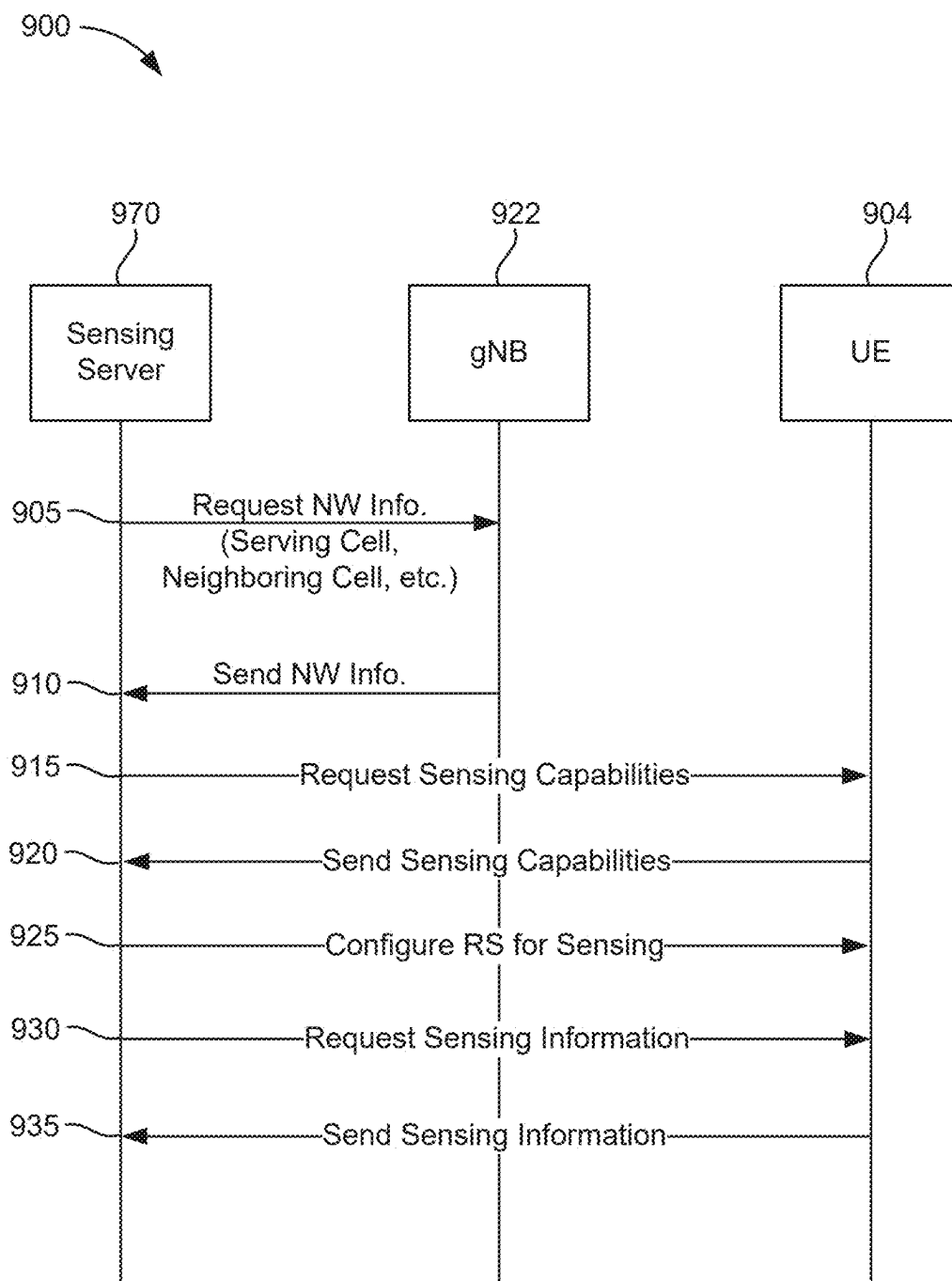


FIG. 9

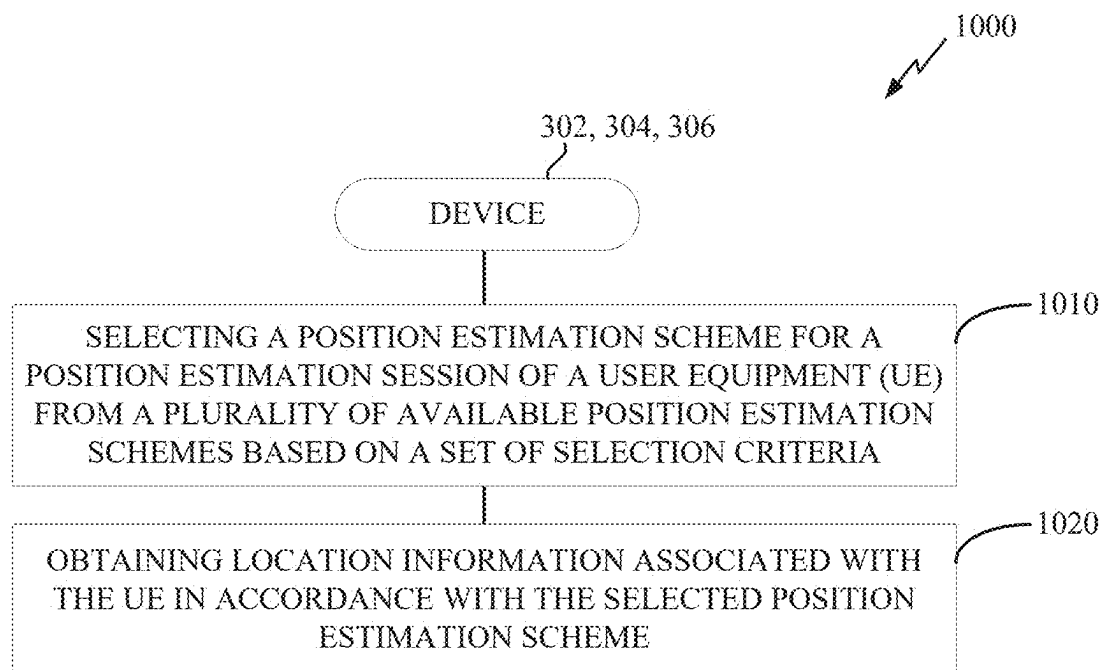


FIG. 10

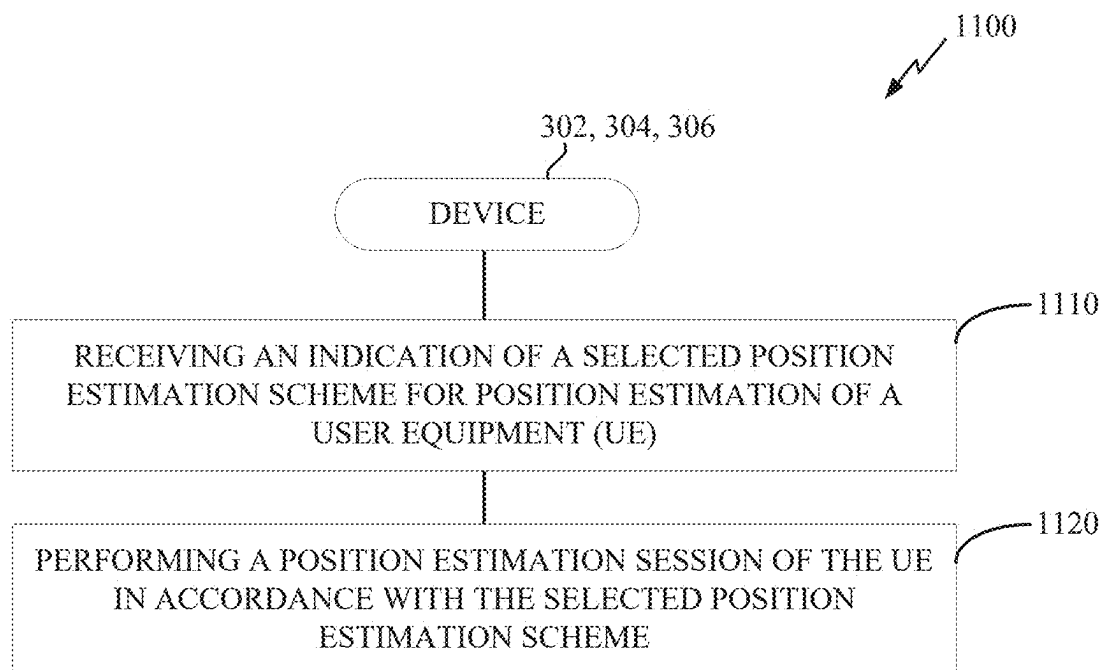


FIG. 11

POSITION ESTIMATION SELECTION SCHEME FOR POSITION ESTIMATION OF A USER EQUIPMENT

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0001] Aspects of the disclosure relate generally to wireless technologies.

2. Description of the Related Art

[0002] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.

[0003] A fifth generation (5G) wireless standard, referred to as New Radio (NR), enables higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide higher data rates as compared to previous standards, more accurate positioning (e.g., based on reference signals for positioning (RS-P), such as downlink, uplink, or sidelink positioning reference signals (PRS)), and other technical enhancements. These enhancements, as well as the use of higher frequency bands, advances in PRS processes and technology, and high-density deployments for 5G, enable highly accurate 5G-based positioning.

SUMMARY

[0004] The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

[0005] In an aspect, a method of operating a device includes selecting a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more

wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtaining location information associated with the UE in accordance with the selected position estimation scheme.

[0006] In an aspect, a method of operating a device includes receiving an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and performing a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0007] In an aspect, a device includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: select a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are

transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtain location information associated with the UE in accordance with the selected position estimation scheme.

[0008] In an aspect, a device includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: receive, via the one or more transceivers, an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and perform a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0009] In an aspect, a device includes means for selecting a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measure-

ment information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and means for obtaining location information associated with the UE in accordance with the selected position estimation scheme.

[0010] In an aspect, a device includes means for receiving an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and means for performing a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0011] In an aspect, a non-transitory computer-readable medium storing computer-executable instructions that, when executed by a device, cause the device to: select a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted

and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtain location information associated with the UE in accordance with the selected position estimation scheme.

[0012] In an aspect, a non-transitory computer-readable medium storing computer-executable instructions that, when executed by a device, cause the device to: receive an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and perform a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0013] Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.

[0015] FIG. 1 illustrates an example wireless communications system, according to aspects of the disclosure.

[0016] FIGS. 2A, 2B, and 2C illustrate example wireless network structures, according to aspects of the disclosure.

[0017] FIGS. 3A, 3B, and 3C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.

[0018] FIG. 4 is a diagram illustrating an example frame structure, according to aspects of the disclosure.

[0019] FIGS. 5A and 5B are diagrams of example sidelink slot structures with and without feedback resources, according to aspects of the disclosure.

[0020] FIGS. 6A to 6D are diagrams illustrating examples of resource pools for positioning, according to aspects of the disclosure.

[0021] FIG. 7 illustrates examples of various positioning methods supported in New Radio (NR), according to aspects of the disclosure.

[0022] FIGS. 8A and 8B illustrate different types of wireless sensing, according to aspects of the disclosure.

[0023] FIG. 9 illustrates an example call flow for a New Radio (NR)-based sensing procedure in which the network configures the sensing parameters, according to aspects of the disclosure.

[0024] FIG. 10 illustrates an exemplary process of communications according to an aspect of the disclosure.

[0025] FIG. 11 illustrates an exemplary process of communications according to an aspect of the disclosure.

DETAILED DESCRIPTION

[0026] Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.

[0027] Various aspects relate generally to a position estimation selection scheme for position estimation of a user equipment (UE). With the anticipated integration of radio frequency (RF) sensing technology in cellular systems in 5G advanced and/or 6G, positioning may leverage the RF sensing service for enhanced power savings and high localization accuracy. For example, for a user equipment (UE) which is interested in positioning, but also wants to save its power, the UE may theoretically be located using RF sensing (passive positioning) without any transmission or measurement from the UE itself. In other words, the UE itself is the sensing target. However, signaling and/or mechanisms to facilitate such passive positioning of UEs is not currently supported. In some designs, position estimation of UEs as a sensing target may be implemented via passive UE position estimation (e.g., the target UE does not transmit or measure any signal associated with the position estimation) and semi-passive position estimation (e.g., some signals relied upon for position estimation are neither transmitted nor measured by the target UE, while other signals relied upon for the position estimation may be transmitted and/or measured by the target UE).

[0028] Particular aspects of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. Aspects of the disclosure

are directed to a position estimation selection scheme for position estimation of a UE. For example, depending on various factors (e.g., UE battery level, positioning accuracy requirements, etc.), different positioning schemes (e.g., PRS-only or legacy positioning, passive UE position estimation, semi-passive UE position estimation, etc.) may be preferred. In an aspect, a set of selection criteria may be evaluated to select the particular position estimation scheme to be implemented for a particular position estimation session of a UE. Such aspects may provide various technical advantages, such as reducing power consumption at the target UE (e.g., particularly advantageous for RedCap UEs), more accurate UE position estimation, reduced UE position estimation latency, and so on.

[0029] The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

[0030] Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

[0031] Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, the sequence(s) of actions described herein can be considered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

[0032] As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be

mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

[0033] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (eNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0034] The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

[0035] In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

[0036] An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

[0037] FIG. 1 illustrates an example wireless communications system 100, according to aspects of the disclosure. The wireless communications system 100 (which may also be referred to as a wireless wide area network (WWAN)) may include various base stations 102 (labeled “BS”) and various UEs 104. The base stations 102 may include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system 100 corresponds to an LTE network, or gNBs where the wireless communications system 100 corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

[0038] The base stations 102 may collectively form a RAN and interface with a core network 170 (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links 122, and through the core network 170 to one or more location servers 172 (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) 172 may be part of core network 170 or may be external to core network 170. A location server 172 may be integrated with a base station 102. A UE 104 may communicate with a location server 172 directly or indirectly. For example, a UE 104 may communicate with a location server 172 via the base station 102 that is currently serving that UE 104. A UE 104 may also communicate with a location server 172 through another path, such as via an application server (not shown), via another network, such as via a wireless local area network (WLAN) access point (AP) (e.g., AP 150 described below), and so on. For signaling purposes, communication between a UE 104 and a location server 172 may be represented as an indirect connection (e.g., through the core network 170, etc.) or a direct connection (e.g., as shown via direct connection 128), with the intervening nodes (if any) omitted from a signaling diagram for clarity.

[0039] In addition to other functions, the base stations 102 may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering,

integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations 102 may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links 134, which may be wired or wireless.

[0040] The base stations 102 may wirelessly communicate with the UEs 104. Each of the base stations 102 may provide communication coverage for a respective geographic coverage area 110. In an aspect, one or more cells may be supported by a base station 102 in each geographic coverage area 110. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas 110.

[0041] While neighboring macro cell base station 102 geographic coverage areas 110 may partially overlap (e.g., in a handover region), some of the geographic coverage areas 110 may be substantially overlapped by a larger geographic coverage area 110. For example, a small cell base station 102' (labeled “SC” for “small cell”) may have a geographic coverage area 110' that substantially overlaps with the geographic coverage area 110 of one or more macro cell base stations 102. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

[0042] The communication links 120 between the base stations 102 and the UEs 104 may include uplink (also referred to as reverse link) transmissions from a UE 104 to a base station 102 and/or downlink (DL) (also referred to as forward link) transmissions from a base station 102 to a UE 104. The communication links 120 may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links 120 may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

[0043] The wireless communications system 100 may further include a wireless local area network (WLAN) access point (AP) 150 in communication with WLAN stations (STAs) 152 via communication links 154 in an unlicensed frequency spectrum (e.g., 5 GHz). When communicating in an unlicensed frequency spectrum, the WLAN STAs 152 and/or the WLAN AP 150 may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

[0044] The small cell base station 102' may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station 102' may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP 150. The small cell base station 102', employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MULTIFIRE®.

[0045] The wireless communications system 100 may further include a millimeter wave (mmW) base station 180 that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE 182. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station 180 and the UE 182 may utilize beamforming (transmit and/or receive) over a mmW communication link 184 to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations 102 may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

[0046] Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate

antennas add together to increase the radiation in a desired direction, while cancelling to suppress radiation in undesired directions.

[0047] Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

[0048] In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

[0049] Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

[0050] Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is

forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

[0051] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the INTERNATIONAL TELECOMMUNICATION UNION® as a “millimeter wave” band.

[0052] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0053] With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band.

[0054] In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE 104/182 and the cell in which the UE 104/182 either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE 104 and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink

and downlink carriers are typically UE-specific. This means that different UEs 104/182 in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE 104/182 at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

[0055] For example, still referring to FIG. 1, one of the frequencies utilized by the macro cell base stations 102 may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations 102 and/or the mmW base station 180 may be secondary carriers (“SCells”). The simultaneous transmission and/or reception of multiple carriers enables the UE 104/182 to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

[0056] The wireless communications system 100 may further include a UE 164 that may communicate with a macro cell base station 102 over a communication link 120 and/or the mmW base station 180 over a mmW communication link 184. For example, the macro cell base station 102 may support a PCell and one or more SCells for the UE 164 and the mmW base station 180 may support one or more SCells for the UE 164.

[0057] In some cases, the UE 164 and the UE 182 may be capable of sidelink communication. Sidelink-capable UEs (SL-UEs) may communicate with base stations 102 over communication links 120 using the Uu interface (i.e., the air interface between a UE and a base station). SL-UEs (e.g., UE 164, UE 182) may also communicate directly with each other over a wireless sidelink 160 using the PC5 interface (i.e., the air interface between sidelink-capable UEs). A wireless sidelink (or just “sidelink”) is an adaptation of the core cellular (e.g., LTE, NR) standard that allows direct communication between two or more UEs without the communication needing to go through a base station. Sidelink communication may be unicast or multicast, and may be used for device-to-device (D2D) media-sharing, vehicle-to-vehicle (V2V) communication, vehicle-to-everything (V2X) communication (e.g., cellular V2X (cV2X) communication, enhanced V2X (eV2X) communication, etc.), emergency rescue applications, etc. One or more of a group of SL-UEs utilizing sidelink communications may be within the geographic coverage area 110 of a base station 102. Other SL-UEs in such a group may be outside the geographic coverage area 110 of a base station 102 or be otherwise unable to receive transmissions from a base station 102. In some cases, groups of SL-UEs communicating via sidelink communications may utilize a one-to-many (1:M) system in which each SL-UE transmits to every other SL-UE in the group. In some cases, a base station 102 facilitates the scheduling of resources for sidelink communications. In other cases, sidelink communications are carried out between SL-UEs without the involvement of a base station 102.

[0058] In an aspect, the sidelink 160 may operate over a wireless communication medium of interest, which may be shared with other wireless communications between other

vehicles and/or infrastructure access points, as well as other RATs. A “medium” may be composed of one or more time, frequency, and/or space communication resources (e.g., encompassing one or more channels across one or more carriers) associated with wireless communication between one or more transmitter/receiver pairs. In an aspect, the medium of interest may correspond to at least a portion of an unlicensed frequency band shared among various RATs. Although different licensed frequency bands have been reserved for certain communication systems (e.g., by a government entity such as the Federal Communications Commission (FCC) in the United States), these systems, in particular those employing small cell access points, have recently extended operation into unlicensed frequency bands such as the Unlicensed National Information Infrastructure (U-NII) band used by wireless local area network (WLAN) technologies, most notably IEEE 802.11x WLAN technologies generally referred to as “Wi-Fi.” Example systems of this type include different variants of CDMA systems, TDMA systems, FDMA systems, orthogonal FDMA (OFDMA) systems, single-carrier FDMA (SC-FDMA) systems, and so on.

[0059] Note that although FIG. 1 only illustrates two of the UEs as SL-UEs (i.e., UEs 164 and 182), any of the illustrated UEs may be SL-UEs. Further, although only UE 182 was described as being capable of beamforming, any of the illustrated UEs, including UE 164, may be capable of beamforming. Where SL-UEs are capable of beamforming, they may beamform towards each other (i.e., towards other SL-UEs), towards other UEs (e.g., UEs 104), towards base stations (e.g., base stations 102, 180, small cell 102', access point 150), etc. Thus, in some cases, UEs 164 and 182 may utilize beamforming over sidelink 160.

[0060] In the example of FIG. 1, any of the illustrated UEs (shown in FIG. 1 as a single UE 104 for simplicity) may receive signals 124 from one or more Earth orbiting space vehicles (SVs) 112 (e.g., satellites). In an aspect, the SVs 112 may be part of a satellite positioning system that a UE 104 can use as an independent source of location information. A satellite positioning system typically includes a system of transmitters (e.g., SVs 112) positioned to enable receivers (e.g., UEs 104) to determine their location on or above the Earth based, at least in part, on positioning signals (e.g., signals 124) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs 112, transmitters may sometimes be located on ground-based control stations, base stations 102, and/or other UEs 104. A UE 104 may include one or more dedicated receivers specifically designed to receive signals 124 for deriving geo location information from the SVs 112.

[0061] In a satellite positioning system, the use of signals 124 can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integrity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multi-functional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation

system (GAGAN), and/or the like. Thus, as used herein, a satellite positioning system may include any combination of one or more global and/or regional navigation satellites associated with such one or more satellite positioning systems.

[0062] In an aspect, SVs 112 may additionally or alternatively be part of one or more non-terrestrial networks (NTNs). In an NTN, an SV 112 is connected to an earth station (also referred to as a ground station, NTN gateway, or gateway), which in turn is connected to an element in a 5G network, such as a modified base station 102 (without a terrestrial antenna) or a network node in a 5GC. This element would in turn provide access to other elements in the 5G network and ultimately to entities external to the 5G network, such as Internet web servers and other user devices. In that way, a UE 104 may receive communication signals (e.g., signals 124) from an SV 112 instead of, or in addition to, communication signals from a terrestrial base station 102.

[0063] The wireless communications system 100 may further include one or more UEs, such as UE 190, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. 1, UE 190 has a D2D P2P link 192 with one of the UEs 104 connected to one of the base stations 102 (e.g., through which UE 190 may indirectly obtain cellular connectivity) and a D2D P2P link 194 with WLAN STA 152 connected to the WLAN AP 150 (through which UE 190 may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links 192 and 194 may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WI-FI DIRECT®, BLUETOOTH®, and so on.

[0064] FIG. 2A illustrates an example wireless network structure 200. For example, a 5GC 210 (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions 214 (e.g., UE registration, authentication, network access, gateway selection, etc.) and user plane (U-plane) functions 212, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) 213 and control plane interface (NG-C) 215 connect the gNB 222 to the 5GC 210 and specifically to the user plane functions 212 and control plane functions 214, respectively. In an additional configuration, an ng-eNB 224 may also be connected to the 5GC 210 via NG-C 215 to the control plane functions 214 and NG-U 213 to user plane functions 212. Further, ng-eNB 224 may directly communicate with gNB 222 via a backhaul connection 223. In some configurations, a Next Generation RAN (NG-RAN) 220 may have one or more gNBs 222, while other configurations include one or more of both ng-eNBs 224 and gNBs 222. Either (or both) gNB 222 or ng-eNB 224 may communicate with one or more UEs 204 (e.g., any of the UEs described herein).

[0065] Another optional aspect may include a location server 230, which may be in communication with the 5GC 210 to provide location assistance for UE(s) 204. The location server 230 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server 230 can be configured to support one or more

location services for UEs 204 that can connect to the location server 230 via the core network, 5GC 210, and/or via the Internet (not illustrated). Further, the location server 230 may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

[0066] FIG. 2B illustrates another example wireless network structure 240. A 5GC 260 (which may correspond to 5GC 210 in FIG. 2A) can be viewed functionally as control plane functions, provided by an access and mobility management function (AMF) 264, and user plane functions, provided by a user plane function (UPF) 262, which operate cooperatively to form the core network (i.e., 5GC 260). The functions of the AMF 264 include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs 204 (e.g., any of the UEs described herein) and a session management function (SMF) 266, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE 204 and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF 264 also interacts with an authentication server function (AUSF) (not shown) and the UE 204, and receives the intermediate key that was established as a result of the UE 204 authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF 264 retrieves the security material from the AUSF. The functions of the AMF 264 also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF 264 also includes location services management for regulatory services, transport for location services messages between the UE 204 and a location management function (LMF) 270 (which acts as a location server 230), transport for location services messages between the NG-RAN 220 and the LMF 270, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE 204 mobility event notification. In addition, the AMF 264 also supports functionalities for non-3GPP® (Third Generation Partnership Project) access networks.

[0067] Functions of the UPF 262 include acting as an anchor point for intra/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QoS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification (service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering, and sending and forwarding of one or more “end markers” to the source RAN node. The UPF 262 may also support transfer of location services messages over a user plane between the UE 204 and a location server, such as an SLP 272.

[0068] The functions of the SMF 266 include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF 262 to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF 266 communicates with the AMF 264 is referred to as the N11 interface.

[0069] Another optional aspect may include an LMF 270, which may be in communication with the 5GC 260 to provide location assistance for UEs 204. The LMF 270 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF 270 can be configured to support one or more location services for UEs 204 that can connect to the LMF 270 via the core network, 5GC 260, and/or via the Internet (not illustrated). The SLP 272 may support similar functions to the LMF 270, but whereas the LMF 270 may communicate with the AMF 264, NG-RAN 220, and UEs 204 over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP 272 may communicate with UEs 204 and external clients (e.g., third-party server 274) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

[0070] Yet another optional aspect may include a third-party server 274, which may be in communication with the LMF 270, the SLP 272, the 5GC 260 (e.g., via the AMF 264 and/or the UPF 262), the NG-RAN 220, and/or the UE 204 to obtain location information (e.g., a location estimate) for the UE 204. As such, in some cases, the third-party server 274 may be referred to as a location services (LCS) client or an external client. The third-party server 274 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server.

[0071] User plane interface 263 and control plane interface 265 connect the 5GC 260, and specifically the UPF 262 and AMF 264, respectively, to one or more gNBs 222 and/or ng-eNBs 224 in the NG-RAN 220. The interface between gNB(s) 222 and/or ng-eNB(s) 224 and the AMF 264 is referred to as the “N2” interface, and the interface between gNB(s) 222 and/or ng-eNB(s) 224 and the UPF 262 is referred to as the “N3” interface. The gNB(s) 222 and/or ng-eNB(s) 224 of the NG-RAN 220 may communicate directly with each other via backhaul connections 223, referred to as the “Xn-C” interface. One or more of gNBs 222 and/or ng-eNBs 224 may communicate with one or more UEs 204 over a wireless interface, referred to as the “Uu” interface.

[0072] The functionality of a gNB 222 may be divided between a gNB central unit (gNB-CU) 226, one or more gNB distributed units (gNB-DUs) 228, and one or more gNB radio units (gNB-RUs) 229. A gNB-CU 226 is a logical node that includes the base station functions of transferring user data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) 228. More specifically, the gNB-CU 226 generally host the radio

resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB 222. A gNB-DU 228 is a logical node that generally hosts the radio link control (RLC) and medium access control (MAC) layer of the gNB 222. Its operation is controlled by the gNB-CU 226. One gNB-DU 228 can support one or more cells, and one cell is supported by only one gNB-DU 228. The interface 232 between the gNB-CU 226 and the one or more gNB-DUs 228 is referred to as the “F1” interface. The physical (PHY) layer functionality of a gNB 222 is generally hosted by one or more standalone gNB-RUs 229 that perform functions such as power amplification and signal transmission/reception. The interface between a gNB-DU 228 and a gNB-RU 229 is referred to as the “Fx” interface. Thus, a UE 204 communicates with the gNB-CU 226 via the RRC, SDAP, and PDCP layers, with a gNB-DU 228 via the RLC and MAC layers, and with a gNB-RU 229 via the PHY layer.

[0073] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, or a network equipment, such as a base station, or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), evolved NB (eNB), NR base station, 5G NB, AP, TRP, cell, etc.) may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station.

[0074] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0075] Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN ALLIANCE®)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0076] FIG. 2C illustrates an example disaggregated base station architecture 250, according to aspects of the disclosure. The disaggregated base station architecture 250 may include one or more central units (CUs) 280 (e.g., gNB-CU 226) that can communicate directly with a core network 267 (e.g., 5GC 210, 5GC 260) via a backhaul link, or indirectly with the core network 267 through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) 259 via an E2 link, or a Non-Real Time (Non-RT) RIC 257 associated with a Service Management and Orchestration (SMO) Framework 255, or both). A CU 280 may communicate with one or more DUs 285 (e.g., gNB-DUs 228) via respective midhaul links, such as an F1 interface. The DUs 285 may communicate with one or more radio units (RUs) 287 (e.g., gNB-RUs 229) via respective fronthaul links. The RUs 287 may communicate with respective UEs 204 via one or more radio frequency (RF) access links. In some implementations, the UE 204 may be simultaneously served by multiple RUs 287.

[0077] Each of the units, i.e., the CUs 280, the DUs 285, the RUs 287, as well as the Near-RT RICs 259, the Non-RT RICs 257 and the SMO Framework 255, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0078] In some aspects, the CU 280 may host one or more higher layer control functions. Such control functions can include RRC, PDCP, service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 280. The CU 280 may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU 280 can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 280 can be implemented to communicate with the DU 285, as necessary, for network control and signaling.

[0079] The DU 285 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 287. In some aspects, the DU 285 may host one or more of a RLC layer, a MAC layer, and one or more high PHY layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP®). In some aspects, the DU 285 may further host one or more low PHY

layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 285, or with the control functions hosted by the CU 280.

[0080] Lower-layer functionality can be implemented by one or more RUs 287. In some deployments, an RU 287, controlled by a DU 285, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) 287 can be implemented to handle over the air (OTA) communication with one or more UEs 204. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 287 can be controlled by the corresponding DU 285. In some scenarios, this configuration can enable the DU(s) 285 and the CU 280 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0081] The SMO Framework 255 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 255 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 255 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) 269) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 280, DUs 285, RUs 287 and Near-RT RICs 259. In some implementations, the SMO Framework 255 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 261, via an O1 interface. Additionally, in some implementations, the SMO Framework 255 can communicate directly with one or more RUs 287 via an O1 interface. The SMO Framework 255 also may include a Non-RT RIC 257 configured to support functionality of the SMO Framework 255.

[0082] The Non-RT RIC 257 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence/machine learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 259. The Non-RT RIC 257 may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC 259. The Near-RT RIC 259 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 280, one or more DUs 285, or both, as well as an O-eNB, with the Near-RT RIC 259.

[0083] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 259, the Non-RT RIC 257 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC 259 and may be received at the SMO Framework 255 or the Non-RT RIC 257 from non-

network data sources or from network functions. In some examples, the Non-RT RIC 257 or the Near-RT RIC 259 may be configured to tune RAN behavior or performance. For example, the Non-RT RIC 257 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 255 (such as reconfiguration via 01) or via creation of RAN management policies (such as A1 policies).

[0084] FIGS. 3A, 3B, and 3C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE 302 (which may correspond to any of the UEs described herein), a base station 304 (which may correspond to any of the base stations described herein), and a network entity 306 (which may correspond to or embody any of the network functions described herein, including the location server 230 and the LMF 270, or alternatively may be independent from the NG-RAN 220 and/or 5GC 210/260 infrastructure depicted in FIGS. 2A and 2B, such as a private network) to support the operations described herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

[0085] The UE 302 and the base station 304 each include one or more wireless wide area network (WWAN) transceivers 310 and 350, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers 310 and 350 may each be connected to one or more antennas 316 and 356, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers 310 and 350 may be variably configured for transmitting and encoding signals 318 and 358 (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals 318 and 358 (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers 310 and 350 include one or more transmitters 314 and 354, respectively, for transmitting and encoding signals 318 and 358, respectively, and one or more receivers 312 and 352, respectively, for receiving and decoding signals 318 and 358, respectively.

[0086] The UE 302 and the base station 304 each also include, at least in some cases, one or more short-range wireless transceivers 320 and 360, respectively. The short-range wireless transceivers 320 and 360 may be connected to one or more antennas 326 and 366, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means

for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., Wi-Fi, LTE Direct, BLUETOOTH®, ZIGBEE®, Z-WAVE®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), ultra-wideband (UWB), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively. As specific examples, the short-range wireless transceivers **320** and **360** may be Wi-Fi transceivers, BLUETOOTH® transceivers, ZIGBEE® and/or Z-WAVE® transceivers, NFC transceivers, UWB transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

[0087] The UE **302** and the base station **304** also include, at least in some cases, satellite signal interfaces **330** and **370**, which each include one or more satellite signal receivers **332** and **372**, respectively, and may optionally include one or more satellite signal transmitters **334** and **374**, respectively. In some cases, the base station **304** may be a terrestrial base station that may communicate with space vehicles (e.g., space vehicles **112**) via the satellite signal interface **370**. In other cases, the base station **304** may be a space vehicle (or other non-terrestrial entity) that uses the satellite signal interface **370** to communicate with terrestrial networks and/or other space vehicles.

[0088] The satellite signal receivers **332** and **372** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal receiver(s) **332** and **372** are satellite positioning system receivers, the satellite positioning/communication signals **338** and **378** may be global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS) signals, etc. Where the satellite signal receiver(s) **332** and **372** are non-terrestrial network (NTN) receivers, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal receiver(s) **332** and **372** may comprise any suitable hardware and/or software for receiving and processing satellite positioning/communication signals **338** and **378**, respectively. The satellite signal receiver(s) **332** and **372** may request information and operations as appropriate from the other systems, and, at least in some cases, perform calculations to determine locations of the UE **302** and the base station **304**, respectively, using measurements obtained by any suitable satellite positioning system algorithm.

[0089] The optional satellite signal transmitter(s) **334** and **374**, when present, may be connected to the one or more

antennas **336** and **376**, respectively, and may provide means for transmitting satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal transmitter(s) **374** are satellite positioning system transmitters, the satellite positioning/communication signals **378** may be GPS signals, GLONASS® signals, Galileo signals, Beidou signals, NAVIC, QZSS signals, etc. Where the satellite signal transmitter(s) **334** and **374** are NTN transmitters, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal transmitter(s) **334** and **374** may comprise any suitable hardware and/or software for transmitting satellite positioning/communication signals **338** and **378**, respectively. The satellite signal transmitter(s) **334** and **374** may request information and operations as appropriate from the other systems.

[0090] The base station **304** and the network entity **306** each include one or more network transceivers **380** and **390**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations **304**, other network entities **306**). For example, the base station **304** may employ the one or more network transceivers **380** to communicate with other base stations **304** or network entities **306** over one or more wired or wireless backhaul links. As another example, the network entity **306** may employ the one or more network transceivers **390** to communicate with one or more base station **304** over one or more wired or wireless backhaul links, or with other network entities **306** over one or more wired or wireless core network interfaces.

[0091] A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) and receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers **380** and **390** in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform transmit “beamforming,” as described herein. Similarly, wireless receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN transceivers **310** and **350**, short-range wireless transceivers **320** and **360**) may also include a network listen module (NLM) or the like for performing various measurements.

[0092] As used herein, the various wireless transceivers (e.g., transceivers 310, 320, 350, and 360, and network transceivers 380 and 390 in some implementations) and wired transceivers (e.g., network transceivers 380 and 390 in some implementations) may generally be characterized as “a transceiver,” “at least one transceiver,” or “one or more transceivers.” As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE 302) and a base station (e.g., base station 304) will generally relate to signaling via a wireless transceiver.

[0093] The UE 302, the base station 304, and the network entity 306 also include other components that may be used in conjunction with the operations as disclosed herein. The UE 302, the base station 304, and the network entity 306 include one or more processors 342, 384, and 394, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors 342, 384, and 394 may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors 342, 384, and 394 may include, for example, one or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

[0094] The UE 302, the base station 304, and the network entity 306 include memory circuitry implementing memories 340, 386, and 396 (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories 340, 386, and 396 may therefore provide means for storing, means for retrieving, means for maintaining, etc. In some cases, the UE 302, the base station 304, and the network entity 306 may include sensing and/or positioning component 348, 388, and 398, respectively. The sensing and/or positioning component 348, 388, and 398 may be hardware circuits that are part of or coupled to the processors 342, 384, and 394, respectively, that, when executed, cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. In other aspects, the sensing and/or positioning component 348, 388, and 398 may be external to the processors 342, 384, and 394 (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the sensing and/or positioning component 348, 388, and 398 may be memory modules stored in the memories 340, 386, and 396, respectively, that, when executed by the processors 342, 384, and 394 (or a modem processing system, another processing system, etc.), cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. FIG. 3A illustrates possible locations of the sensing and/or positioning component 348, which may be, for example, part of the one or more WWAN transceivers 310, the memory 340, the one or more processors 342, or any combination thereof, or may be a standalone component. FIG. 3B illustrates possible locations of the sensing and/or positioning component 388, which may be, for example, part of the one or more WWAN

transceivers 350, the memory 386, the one or more processors 384, or any combination thereof, or may be a standalone component. FIG. 3C illustrates possible locations of the sensing and/or positioning component 398, which may be, for example, part of the one or more network transceivers 390, the memory 396, the one or more processors 394, or any combination thereof, or may be a standalone component.

[0095] The UE 302 may include one or more sensors 344 coupled to the one or more processors 342 to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers 310, the one or more short-range wireless transceivers 320, and/or the satellite signal interface 330. By way of example, the sensor(s) 344 may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) 344 may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) 344 may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems.

[0096] In addition, the UE 302 includes a user interface 346 providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of a sensing device such as a keypad, a touch screen, a microphone, and so on). Although not shown, the base station 304 and the network entity 306 may also include user interfaces.

[0097] Referring to the one or more processors 384 in more detail, in the downlink, IP packets from the network entity 306 may be provided to the processor 384. The one or more processors 384 may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors 384 may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error correction, priority handling, and logical channel prioritization.

[0098] The transmitter 354 and the receiver 352 may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the

transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter 354 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 302. Each spatial stream may then be provided to one or more different antennas 356. The transmitter 354 may modulate an RF carrier with a respective spatial stream for transmission.

[0099] At the UE 302, the receiver 312 receives a signal through its respective antenna(s) 316. The receiver 312 recovers information modulated onto an RF carrier and provides the information to the one or more processors 342. The transmitter 314 and the receiver 312 implement Layer-1 functionality associated with various signal processing functions. The receiver 312 may perform spatial processing on the information to recover any spatial streams destined for the UE 302. If multiple spatial streams are destined for the UE 302, they may be combined by the receiver 312 into a single OFDM symbol stream. The receiver 312 then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station 304. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station 304 on the physical channel. The data and control signals are then provided to the one or more processors 342, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

[0100] In the downlink, the one or more processors 342 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors 342 are also responsible for error detection.

[0101] Similar to the functionality described in connection with the downlink transmission by the base station 304, the one or more processors 342 provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer func-

tionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel prioritization.

[0102] Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station 304 may be used by the transmitter 314 to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter 314 may be provided to different antenna(s) 316. The transmitter 314 may modulate an RF carrier with a respective spatial stream for transmission.

[0103] The uplink transmission is processed at the base station 304 in a manner similar to that described in connection with the receiver function at the UE 302. The receiver 352 receives a signal through its respective antenna(s) 356. The receiver 352 recovers information modulated onto an RF carrier and provides the information to the one or more processors 384.

[0104] In the uplink, the one or more processors 384 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE 302. IP packets from the one or more processors 384 may be provided to the core network. The one or more processors 384 are also responsible for error detection.

[0105] For convenience, the UE 302, the base station 304, and/or the network entity 306 are shown in FIGS. 3A, 3B, and 3C as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In particular, various components in FIGS. 3A to 3C are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. 3A, a particular implementation of UE 302 may omit the WWAN transceiver(s) 310 (e.g., a wearable device or tablet computer or personal computer (PC) or laptop may have Wi-Fi and/or BLUETOOTH® capability without cellular capability), or may omit the short-range wireless transceiver(s) 320 (e.g., cellular-only, etc.), or may omit the satellite signal interface 330, or may omit the sensor(s) 344, and so on. In another example, in case of FIG. 3B, a particular implementation of the base station 304 may omit the WWAN transceiver(s) 350 (e.g., a Wi-Fi “hotspot” access point without cellular capability), or may omit the short-range wireless transceiver(s) 360 (e.g., cellular-only, etc.), or may omit the satellite signal interface 370, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

[0106] The various components of the UE 302, the base station 304, and the network entity 306 may be communicatively coupled to each other over data buses 308, 382, and 392, respectively. In an aspect, the data buses 308, 382, and 392 may form, or be part of, a communication interface of the UE 302, the base station 304, and the network entity 306,

respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station 304), the data buses 308, 382, and 392 may provide communication between them.

[0107] The components of FIGS. 3A, 3B, and 3C may be implemented in various ways. In some implementations, the components of FIGS. 3A, 3B, and 3C may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks 310 to 346 may be implemented by processor and memory component(s) of the UE 302 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks 350 to 388 may be implemented by processor and memory component(s) of the base station 304 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks 390 to 398 may be implemented by processor and memory component(s) of the network entity 306 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed “by a UE,” “by a base station,” “by a network entity,” etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE 302, base station 304, network entity 306, etc., such as the processors 342, 384, 394, the transceivers 310, 320, 350, and 360, the memories 340, 386, and 396, the sensing and/or positioning component 348, 388, and 398, etc.

[0108] In some designs, the network entity 306 may be implemented as a core network component. In other designs, the network entity 306 may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN 220 and/or 5GC 210/260). For example, the network entity 306 may be a component of a private network that may be configured to communicate with the UE 302 via the base station 304 or independently from the base station 304 (e.g., over a non-cellular communication link, such as Wi-Fi).

[0109] Note that the UE 302 illustrated in FIG. 3A may represent a “reduced capability” (“RedCap”) UE or a “premium” UE. As described further below, while RedCap and premium UEs may have the same types of components (e.g., both may have one or more WWAN transceivers 310, one or more short-range wireless transceivers 320, satellite signal interface 330, one or more processors 342, memory 340, etc.), the components may have different degrees of functionality (e.g., increased or decreased performance, more or fewer capabilities, etc.) depending on whether the UE 302 corresponds to a RedCap UE or a premium UE.

[0110] UEs may be classified as RedCap UEs (e.g., wearables, such as smart watches, glasses, rings, etc.) and premium UEs (e.g., smartphones, tablet computers, laptop computers, etc.). RedCap UEs may alternatively be referred to as low-tier UEs, light UEs, or super light UEs. Premium UEs may alternatively be referred to as full-capability UEs or simply UEs. RedCap UEs generally have lower baseband

processing capability, fewer antennas (e.g., one receiver antenna as baseline in FR1 or FR2, two receiver antennas optionally), lower operational bandwidth capabilities (e.g., 20 MHz for FR1 with no supplemental uplink or carrier aggregation, or 50 or 100 MHz for FR2), only half duplex frequency division duplex (HD-FDD) capability, smaller HARQ buffer, reduced physical downlink control channel (PDCCH) monitoring, restricted modulation (e.g., 64 QAM for downlink and 16 QAM for uplink), relaxed processing timeline requirements, and/or lower uplink transmission power compared to premium UEs. Different UE tiers can be differentiated by UE category and/or by UE capability. For example, certain types of UEs may be assigned a classification (e.g., by the original equipment manufacturer (OEM), the applicable wireless communications standards, or the like) of “RedCap” and other types of UEs may be assigned a classification of “premium.” Certain tiers of UEs may also report their type (e.g., “RedCap” or “premium”) to the network. Additionally, certain resources and/or channels may be dedicated to certain types of UEs.

[0111] As will be appreciated, the accuracy of RedCap UE positioning may be limited. For example, a RedCap UE may operate on a reduced bandwidth, such as 5 to 20 MHz for wearable devices and “relaxed” IoT devices (i.e., IoT devices with relaxed, or lower, capability parameters, such as lower throughput, relaxed delay requirements, lower energy consumption, etc.), which results in lower positioning accuracy. As another example, a RedCap UE’s receive processing capability may be limited due to its lower cost RF/baseband. As such, the reliability of measurements and positioning computations would be reduced. In addition, such a RedCap UE may not be able to receive multiple PRS from multiple TRPs, further reducing positioning accuracy. As yet another example, the transmit power of a RedCap UE may be reduced, meaning there would be a lower quality of uplink measurements for RedCap UE positioning.

[0112] Premium UEs generally have a larger form factor and are costlier than RedCap UEs, and have more features and capabilities than RedCap UEs. For example, with respect to positioning, a premium UE may operate on the full PRS bandwidth, such as 100 MHz, and measure PRS from more TRPs than RedCap UEs, both of which result in higher positioning accuracy. As another example, a premium UE’s receive processing capability may be higher (e.g., faster) due to its higher-capability RF/baseband. In addition, the transmit power of a premium UE may be higher than that of a RedCap UE. As such, the reliability of measurements and positioning computations would be increased.

[0113] Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. 4 is a diagram 400 illustrating an example frame structure, according to aspects of the disclosure. The frame structure may be a downlink or uplink frame structure. Other wireless communications technologies may have different frame structures and/or different channels.

[0114] LTE, and in some cases NR, utilizes orthogonal frequency-division multiplexing (OFDM) on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with

data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal fast Fourier transform (FFT) size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., 6 resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHz, respectively.

[0115] LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz ($\mu=0$), 30 kHz ($\mu=1$), 60 kHz ($\mu=2$), 120 kHz ($\mu=3$), and 240 kHz ($\mu=4$) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS ($\mu=0$), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (us), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS ($\mu=1$), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 us, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS ($\mu=2$), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is 16.7 us, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS ($\mu=3$), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 us, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS ($\mu=4$), there are 16 slots per subframe, 160 slots per frame, the slot duration is 0.0625 ms, the symbol duration is 4.17 us, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.

[0116] In the example of FIG. 4, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIG. 4, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.

[0117] A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the numerology of FIG. 4, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.

[0118] Some of the REs may carry reference (pilot) signals (RS). The reference signals may include positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), sounding reference signals (SRS), etc., depending on whether the illustrated frame structure is used for uplink or downlink communication. FIG. 4 illustrates example locations of REs carrying a reference signal (labeled “R”).

[0119] A collection of resource elements (REs) that are used for transmission of PRS is referred to as a “PRS resource.” The collection of resource elements can span multiple PRBs in the frequency domain and ‘N’ (such as 1 or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol in the time domain, a PRS resource occupies consecutive PRBs in the frequency domain.

[0120] The transmission of a PRS resource within a given PRB has a particular comb size (also referred to as the “comb density”). A comb size ‘N’ represents the subcarrier spacing (or frequency/tone spacing) within each symbol of a PRS resource configuration. Specifically, for a comb size ‘N,’ PRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the PRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit PRS of the PRS resource. Currently, comb sizes of comb-2, comb-4, comb-6, and comb-12 are supported for DL-PRS. FIG. 4 illustrates an example PRS resource configuration for comb-4 (which spans four symbols). That is, the locations of the shaded REs (labeled “R”) indicate a comb-4 PRS resource configuration.

[0121] Currently, a DL-PRS resource may span 2, 4, 6, or 12 consecutive symbols within a slot with a fully frequency-domain staggered pattern. A DL-PRS resource can be configured in any higher layer configured downlink or flexible (FL) symbol of a slot. There may be a constant energy per resource element (EPRE) for all REs of a given DL-PRS resource. The following are the frequency offsets from symbol to symbol for comb sizes 2, 4, 6, and 12 over 2, 4, 6, and 12 symbols. 2-symbol comb-2: {0, 1}; 4-symbol comb-2: {0, 1, 0, 1}; 6-symbol comb-2: {0, 1, 0, 1, 0, 1}; 12-symbol comb-2: {0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1}; 4-symbol comb-4: {0, 2, 1, 3} (as in the example of FIG. 4); 12-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3}; 6-symbol comb-6: {0, 3, 1, 4, 2, 5}; 12-symbol comb-6: {0, 3, 1, 4, 2, 5, 0, 3, 1, 4, 2, 5}; and 12-symbol comb-12: {0, 6, 3, 9, 1, 7, 4, 10, 2, 8, 5, 11}.

[0122] A “PRS resource set” is a set of PRS resources used for the transmission of PRS signals, where each PRS resource has a PRS resource ID. In addition, the PRS resources in a PRS resource set are associated with the same TRP. A PRS resource set is identified by a PRS resource set ID and is associated with a particular TRP (identified by a TRP ID). In addition, the PRS resources in a PRS resource set have the same periodicity, a common muting pattern configuration, and the same repetition factor (such as “PRS-ResourceRepetitionFactor”) across slots. The periodicity is the time from the first repetition of the first PRS resource of a first PRS instance to the same first repetition of the same first PRS resource of the next PRS instance. The periodicity

may have a length selected from $2^{\mu} \cdot \{4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240\}$ slots, with $\mu=0, 1, 2, 3$. The repetition factor may have a length selected from $\{1, 2, 4, 6, 8, 16, 32\}$ slots.

[0123] A PRS resource ID in a PRS resource set is associated with a single beam (or beam ID) transmitted from a single TRP (where a TRP may transmit one or more beams). That is, each PRS resource of a PRS resource set may be transmitted on a different beam, and as such, a “PRS resource,” or simply “resource,” also can be referred to as a “beam.” Note that this does not have any implications on whether the TRPs and the beams on which PRS are transmitted are known to the UE.

[0124] A “PRS instance” or “PRS occasion” is one instance of a periodically repeated time window (such as a group of one or more consecutive slots) where PRS are expected to be transmitted. A PRS occasion also may be referred to as a “PRS positioning occasion,” a “PRS positioning instance,” a “positioning occasion,” a “positioning instance,” a “positioning repetition,” or simply an “occasion,” an “instance,” or a “repetition.”

[0125] A “positioning frequency layer” (also referred to simply as a “frequency layer”) is a collection of one or more PRS resource sets across one or more TRPs that have the same values for certain parameters. Specifically, the collection of PRS resource sets has the same subcarrier spacing and cyclic prefix (CP) type (meaning all numerologies supported for the physical downlink shared channel (PDSCH) are also supported for PRS), the same Point A, the same value of the downlink PRS bandwidth, the same start PRB (and center frequency), and the same comb-size. The Point A parameter takes the value of the parameter “ARFCN-ValueNR” (where “ARFCN” stands for “absolute radio-frequency channel number”) and is an identifier/code that specifies a pair of physical radio channel used for transmission and reception. The downlink PRS bandwidth may have a granularity of four PRBs, with a minimum of 24 PRBs and a maximum of 272 PRBs. Currently, up to four frequency layers have been defined, and up to two PRS resource sets may be configured per TRP per frequency layer.

[0126] The concept of a frequency layer is somewhat like the concept of component carriers and bandwidth parts (BWPs), but different in that component carriers and BWPs are used by one base station (or a macro cell base station and a small cell base station) to transmit data channels, while frequency layers are used by several (usually three or more) base stations to transmit PRS. A UE may indicate the number of frequency layers it can support when it sends the network its positioning capabilities, such as during an LTE positioning protocol (LPP) session. For example, a UE may indicate whether it can support one or four positioning frequency layers.

[0127] Note that the terms “positioning reference signal” and “PRS” generally refer to specific reference signals that are used for positioning in NR and LTE systems. However, as used herein, the terms “positioning reference signal” and “PRS” may also refer to any type of reference signal that can be used for positioning, such as but not limited to, PRS as defined in LTE and NR, TRS, PTRS, CRS, CSI-RS, DMRS, PSS, SSS, SSB, SRS, UL-PRS, etc. In addition, the terms “positioning reference signal” and “PRS” may refer to downlink, uplink, or sidelink positioning reference signals, unless otherwise indicated by the context. If needed to

further distinguish the type of PRS, a downlink positioning reference signal may be referred to as a “DL-PRS,” an uplink positioning reference signal (e.g., an SRS-for-positioning, PTRS) may be referred to as an “UL-PRS,” and a sidelink positioning reference signal may be referred to as an “SL-PRS.” In addition, for signals that may be transmitted in the downlink, uplink, and/or sidelink (e.g., DMRS), the signals may be prepended with “DL,” “UL,” or “SL” to distinguish the direction. For example, “UL-DMRS” is different from “DL-DMRS.”

[0128] In an aspect, the reference signal carried on the REs labeled “R” in FIG. 4 may be SRS. SRS transmitted by a UE may be used by a base station to obtain the channel state information (CSI) for the transmitting UE. CSI describes how an RF signal propagates from the UE to the base station and represents the combined effect of scattering, fading, and power decay with distance. The system uses the SRS for resource scheduling, link adaptation, massive MIMO, beam management, etc.

[0129] A collection of REs that are used for transmission of SRS is referred to as an “SRS resource,” and may be identified by the parameter “SRS-ResourceId.” The collection of resource elements can span multiple PRBs in the frequency domain and ‘N’ (e.g., one or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol, an SRS resource occupies one or more consecutive PRBs. An “SRS resource set” is a set of SRS resources used for the transmission of SRS signals, and is identified by an SRS resource set ID (“SRS-Resource-SetId”).

[0130] The transmission of SRS resources within a given PRB has a particular comb size (also referred to as the “comb density”). A comb size ‘N’ represents the subcarrier spacing (or frequency/tone spacing) within each symbol of an SRS resource configuration. Specifically, for a comb size ‘N,’ SRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the SRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit SRS of the SRS resource. In the example of FIG. 4, the illustrated SRS is comb-4 over four symbols. That is, the locations of the shaded SRS REs indicate a comb-4 SRS resource configuration.

[0131] Currently, an SRS resource may span 1, 2, 4, 8, or 12 consecutive symbols within a slot with a comb size of comb-2, comb-4, or comb-8. The following are the frequency offsets from symbol to symbol for the SRS comb patterns that are currently supported. 1-symbol comb-2: {0}; 2-symbol comb-2: {0, 1}; 2-symbol comb-4: {0, 2}; 4-symbol comb-2: {0, 1, 0, 1}; 4-symbol comb-4: {0, 2, 1, 3} (as in the example of FIG. 4); 8-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3}; 12-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3}; 4-symbol comb-8: {0, 4, 2, 6}; 8-symbol comb-8: {0, 4, 2, 6, 1, 5, 3, 7}; and 12-symbol comb-8: {0, 4, 2, 6, 1, 5, 3, 7, 0, 4, 2, 6}.

[0132] Generally, as noted above, a UE transmits SRS to enable the receiving base station (either the serving base station or a neighboring base station) to measure the channel quality (i.e., CSI) between the UE and the base station. However, SRS can also be specifically configured as uplink positioning reference signals for uplink-based positioning procedures, such as uplink time difference of arrival (UL-TDOA), round-trip-time (RTT), uplink angle-of-arrival (UL-AoA), etc. As used herein, the term “SRS” may refer to

SRS configured for channel quality measurements or SRS configured for positioning purposes. The former may be referred to herein as “SRS-for-communication” and/or the latter may be referred to as “SRS-for-positioning” or “positioning SRS” when needed to distinguish the two types of SRS.

[0133] Several enhancements over the previous definition of SRS may be available for SRS-for-positioning (also referred to as “UL-PRS”), such as a new staggered pattern within an SRS resource (except for single-symbol/comb-2), a new comb type for SRS, new sequences for SRS, a higher number of SRS resource sets per component carrier, and a higher number of SRS resources per component carrier. In addition, the parameters “SpatialRelationInfo” and “Path-LossReference” are to be configured based on a downlink reference signal or SSB from a neighboring TRP. Further still, one SRS resource may be transmitted outside the active BWP, and one SRS resource may span across multiple component carriers. Also, SRS may be configured in RRC connected state and only transmitted within an active BWP. Further, there may be no frequency hopping, no repetition factor, a single antenna port, and new lengths for SRS (e.g., 8 and 12 symbols). There also may be open-loop power control and not closed-loop power control, and comb-8 (i.e., an SRS transmitted every eighth subcarrier in the same symbol) may be used. Lastly, the UE may transmit through the same transmit beam from multiple SRS resources for UL-AoA. These features may be configured through RRC higher layer signaling (and potentially triggered or activated through a MAC control element (MAC-CE) or downlink control information (DCI)).

[0134] Sidelink communication takes place in transmission or reception resource pools. In the frequency domain, the minimum resource allocation unit is a sub-channel (e.g., a collection of consecutive PRBs in the frequency domain). In the time domain, resource allocation is in one slot intervals. However, some slots are not available for sidelink, and some slots contain feedback resources. In addition, sidelink resources can be (pre) configured to occupy fewer than the 14 symbols of a slot.

[0135] Sidelink resources are configured at the radio resource control (RRC) layer. The RRC configuration can be by pre-configuration (e.g., preloaded on the UE) or configuration (e.g., from a serving base station).

[0136] NR sidelinks support hybrid automatic repeat request (HARQ) retransmission. FIG. 5A is a diagram 500 of an example slot structure without feedback resources, according to aspects of the disclosure. In the example of FIG. 5A, time is represented horizontally and frequency is represented vertically. In the time domain, the length of each block is one orthogonal frequency division multiplexing (OFDM) symbol, and the 14 symbols make up a slot. In the frequency domain, the height of each block is one sub-channel. Currently, the (pre) configured sub-channel size can be selected from the set of {10, 15, 20, 25, 50, 75, 100} physical resource blocks (PRBs).

[0137] For a sidelink slot, the first symbol is a repetition of the preceding symbol and is used for automatic gain control (AGC) setting. This is illustrated in FIG. 5A by the vertical and horizontal hashing. As shown in FIG. 5A, for sidelink, the physical sidelink control channel (PSCCH) and the physical sidelink shared channel (PSSCH) are transmitted in the same slot. Similar to the physical downlink control channel (PDCCH), the PSCCH carries control information

about sidelink resource allocation and descriptions about sidelink data transmitted to the UE. Likewise, similar to the physical downlink shared channel (PDSCH), the PSSCH carries user data for the UE. In the example of FIG. 5A, the PSCCH occupies half the bandwidth of the sub-channel and only three symbols. Finally, a gap symbol is present after the PSSCH.

[0138] FIG. 5B is a diagram 550 of an example slot structure with feedback resources, according to aspects of the disclosure. In the example of FIG. 5B, time is represented horizontally and frequency is represented vertically. In the time domain, the length of each block is one OFDM symbol, and the 14 symbols make up a slot. In the frequency domain, the height of each block is one sub-channel.

[0139] The slot structure illustrated in FIG. 5B is similar to the slot structure illustrated in FIG. 5A, except that the slot structure illustrated in FIG. 5B includes feedback resources. Specifically, two symbols at the end of the slot have been dedicated to the physical sidelink feedback channel (PSFCH). The first PSFCH symbol is a repetition of the second PSFCH symbol for AGC setting. In addition to the gap symbol after the PSSCH, there is a gap symbol after the two PSFCH symbols. Currently, resources for the PSFCH can be configured with a periodicity selected from the set of {0, 1, 2, 4} slots.

[0140] FIG. 6A is a diagram 600 illustrating an example of a resource pool for positioning configured within a sidelink resource pool for communication (i.e., a shared resource pool), according to aspects of the disclosure. In the example of FIG. 6A, time is represented horizontally and frequency is represented vertically. In the time domain, the length of each block is an orthogonal frequency division multiplexing (OFDM) symbol, and the 14 symbols make up a slot. In the frequency domain, the height of each block is a sub-channel.

[0141] In the example of FIG. 6A, the entire slot (except for the first and last symbols) can be a resource pool for sidelink communication. That is, any of the symbols other than the first and last can be allocated for sidelink communication. However, a resource pool for positioning (RP-P) is allocated in the last four pre-gap symbols of the slot. As such, non-sidelink positioning data, such as user data (PSSCH), channel state information reference signal (CSI-RS), and control information, can only be transmitted in the first eight post-automatic gain control (AGC) symbols and not in the last four pre-gap symbols to prevent a collision with the configured RP-P. The non-sidelink positioning data that would otherwise be transmitted in the last four pre-gap symbols can be punctured or muted, or the non-sidelink data that would normally span more than the eight post-AGC symbols can be rate matched to fit into the eight post-AGC symbols.

[0142] Sidelink positioning reference signals (SL-PRS) have been defined to enable sidelink positioning procedures among UEs. Like a downlink PRS (DL-PRS), a SL-PRS resource is composed of one or more resource elements (i.e., one OFDM symbol in the time domain and one subcarrier in the frequency domain). SL-PRS resources have been designed with a comb-based pattern to enable fast Fourier transform (FFT)-based processing at the receiver. SL-PRS resources are composed of unstaggered, or only partially staggered, resource elements in the frequency domain to provide small time of arrival (TOA) uncertainty and reduced overhead of each SL-PRS resource. SL-PRS may also be associated with specific RP-Ps (e.g., certain SL-PRS may be

allocated in certain RP-Ps). SL-PRS have also been defined with intra-slot repetition (not shown in FIG. 6A) to allow for combining gains (if needed). There may also be inter-UE coordination of RP-Ps to provide for dynamic SL-PRS and data multiplexing while minimizing SL-PRS collisions.

[0143] FIGS. 6B and 6C are diagrams 630 and 650, respectively, illustrating additional examples of resource pools for positioning configured within sidelink resource pools for communication. Similar to FIG. 6, the examples of FIGS. 6B and 6C illustrate shared resource pool structures. With respect to FIGS. 6B and 6C, in some designs, the following parameters may be defined, for example: physical sidelink control channel (PSCCH) and SL-PRS are only time-division multiplexed, PSSCH and SL-PRS are only time-division multiplexed (e.g., the maximum comb size is 4), PSSCH carries both type 2 sidelink control information (SCI-2) and a sidelink shared channel (SL-SCH) (e.g., a new SCI-2 format is introduced), SL-PRS is mapped on consecutive symbols, SL-PRS is not mapped on symbols with PSSCH demodulation reference signals (DMRS), and/or SL-PRS transmit power is the same as the transmit power of the PSSCH (e.g., this implies per-resource element power boosting will be applied for comb-2 and comb-4).

[0144] FIG. 6D is a diagram 670 illustrating another example of a resource pool for positioning configured within a sidelink resource pool for communication. In the example of FIG. 6D, a dedicated resource pool structure is depicted. With respect to FIG. 6D, in some designs, the following parameters may be defined, for example: SL-PRS is immediately preceded by an AGC symbol, SL-PRS is immediately followed by a gap symbol (at least when the gap symbol is the last sidelink symbol in a slot), PSCCH and SL-PRS can only be time-division multiplexed, different comb sizes (N) and SL-PRS durations (M) can be supported in the same resource pool (e.g., one set of SL-PRS resources can only have a single (M, N) combination), PSSCH is mapped to the first sidelink symbols in a slot, the number of PSCCH symbols is (pre-) configured to 1, 2, or 3, the number of physical resource blocks is (pre-) configured using sidelink communications values, and/or there is a one-to-one implicit mapping between PSCCH and SL-PRS.

[0145] In some designs, in a shared resource pool, with regards to the fields in SCI format 2-D, the following fields may be included, for example: a SL-PRS resource information indication of the current slot (ceiling ($\log_2$ (#SL-PRS resources (pre-) configured in the resource pool) bits)), SL-PRS request (0 or 1 bit), and/or embedded SCI format ([X] bit(s)). If the “embedded SCI format” field is set to [0], the SCI 2-A fields are included with necessary padding. If the “embedded SCI format” field is set to [1], the SCI 2-B fields are included.

[0146] In some designs, for a shared resource pool, there may be an explicit (pre-) configuration of SL-PRS resources in a slot, applicable for an indicated frequency domain allocation, which includes, for example: SL-PRS Resource ID, (M, N) pattern, and/or comb offset. In some designs, for a given value of ‘M,’ a SL-PRS resource is mapped to the last consecutive ‘M’ sidelink symbol(s) in the slot that can be used for SL-PRS, taking into consideration multiplexing with PSSCH DMRS, phase tracking reference signals (PT-RS), CSI-RS, PSFCH, gap symbols, AGC symbols, and/or PSCCH in the slot. In some designs, the maximum number of SL-PRS resources in a slot of a shared resource pool may be (pre-) configured.

[0147] In some designs, in dedicated resource pools, with regards to the procedure for determining the subset of resources to be reported to higher layers, when triggering the resource (re-) selection procedure, the higher layers provide the following parameters for candidate SL-PRS transmission(s), for example: resource pool from which to report SL-PRS resources, priority, delay budget, reservation period, list of resources for pre-emption and re-evaluation, and/or the set of SL-PRS resource identifiers that can include all (pre-) configured SL-PRS resource identifiers.

[0148] NR supports a number of cellular network-based positioning technologies, including downlink-based, uplink-based, and downlink-and-uplink-based positioning methods. Downlink-based positioning methods include observed time difference of arrival (OTDOA) in LTE, downlink time difference of arrival (DL-TDOA) in NR, and downlink angle-of-departure (DL-AoD) in NR. FIG. 7 illustrates examples of various positioning methods, according to aspects of the disclosure. In an OTDOA or DL-TDOA positioning procedure, illustrated by scenario 710, a UE measures the differences between the times of arrival (ToAs) of reference signals (e.g., positioning reference signals (PRS)) received from pairs of base stations, referred to as reference signal time difference (RSTD) or time difference of arrival (TDOA) measurements, and reports them to a positioning entity. More specifically, the UE receives the identifiers (IDs) of a reference base station (e.g., a serving base station) and multiple non-reference base stations in assistance data. The UE then measures the RSTD between the reference base station and each of the non-reference base stations. Based on the known locations of the involved base stations and the RSTD measurements, the positioning entity (e.g., the UE for UE-based positioning or a location server for UE-assisted positioning) can estimate the UE’s location.

[0149] For DL-AoD positioning, illustrated by scenario 720, the positioning entity uses a measurement report from the UE of received signal strength measurements of multiple downlink transmit beams to determine the angle(s) between the UE and the transmitting base station(s). The positioning entity can then estimate the location of the UE based on the determined angle(s) and the known location(s) of the transmitting base station(s).

[0150] Uplink-based positioning methods include uplink time difference of arrival (UL-TDOA) and uplink angle-of-arrival (UL-AoA). UL-TDOA is similar to DL-TDOA, but is based on uplink reference signals (e.g., sounding reference signals (SRS)) transmitted by the UE to multiple base stations. Specifically, a UE transmits one or more uplink reference signals that are measured by a reference base station and a plurality of non-reference base stations. Each base station then reports the reception time (referred to as the relative time of arrival (RTOA)) of the reference signal(s) to a positioning entity (e.g., a location server) that knows the locations and relative timing of the involved base stations. Based on the reception-to-reception (Rx-Rx) time difference between the reported RTOA of the reference base station and the reported RTOA of each non-reference base station, the known locations of the base stations, and their known timing offsets, the positioning entity can estimate the location of the UE using TDOA.

[0151] For UL-AoA positioning, one or more base stations measure the received signal strength of one or more uplink reference signals (e.g., SRS) received from a UE on one or more uplink receive beams. The positioning entity uses the

signal strength measurements and the angle(s) of the receive beam(s) to determine the angle(s) between the UE and the base station(s). Based on the determined angle(s) and the known location(s) of the base station(s), the positioning entity can then estimate the location of the UE.

[0152] Downlink-and-uplink-based positioning methods include enhanced cell-ID (E-CID) positioning and multi-round-trip-time (RTT) positioning (also referred to as “multi-cell RTT” and “multi-RTT”). In an RTT procedure, a first entity (e.g., a base station or a UE) transmits a first RTT-related signal (e.g., a PRS or SRS) to a second entity (e.g., a UE or base station), which transmits a second RTT-related signal (e.g., an SRS or PRS) back to the first entity. Each entity measures the time difference between the time of arrival (ToA) of the received RTT-related signal and the transmission time of the transmitted RTT-related signal. This time difference is referred to as a reception-to-transmission (Rx-Tx) time difference. The Rx-Tx time difference measurement may be made, or may be adjusted, to include only a time difference between nearest slot boundaries for the received and transmitted signals. Both entities may then send their Rx-Tx time difference measurement to a location server (e.g., an LMF 270), which calculates the round trip propagation time (i.e., RTT) between the two entities from the two Rx-Tx time difference measurements (e.g., as the sum of the two Rx-Tx time difference measurements). Alternatively, one entity may send its Rx-Tx time difference measurement to the other entity, which then calculates the RTT. The distance between the two entities can be determined from the RTT and the known signal speed (e.g., the speed of light). For multi-RTT positioning, illustrated by scenario 730, a first entity (e.g., a UE or base station) performs an RTT positioning procedure with multiple second entities (e.g., multiple base stations or UEs) to enable the location of the first entity to be determined (e.g., using multilateration) based on distances to, and the known locations of, the second entities. RTT and multi-RTT methods can be combined with other positioning techniques, such as UL-AoA and DL-AoD, to improve location accuracy, as illustrated by scenario 740.

[0153] The E-CID positioning method is based on radio resource management (RRM) measurements. In E-CID, the UE reports the serving cell ID, the timing advance (TA), and the identifiers, estimated timing, and signal strength of detected neighbor base stations. The location of the UE is then estimated based on this information and the known locations of the base station(s).

[0154] To assist positioning operations, a location server (e.g., location server 230, LMF 270, SLP 272) may provide assistance data to the UE. For example, the assistance data may include identifiers of the base stations (or the cells/TRPs of the base stations) from which to measure reference signals, the reference signal configuration parameters (e.g., the number of consecutive slots including PRS, periodicity of the consecutive slots including PRS, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth, etc.), and/or other parameters applicable to the particular positioning method. Alternatively, the assistance data may originate directly from the base stations themselves (e.g., in periodically broadcasted overhead messages, etc.). In some cases, the UE may be able to detect neighbor network nodes itself without the use of assistance data.

[0155] In the case of an OTDOA or DL-TDOA positioning procedure, the assistance data may further include an expected RSTD value and an associated uncertainty, or search window, around the expected RSTD. In some cases, the value range of the expected RSTD may be ± 500 microseconds (μ s). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be ± 32 μ s. In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be ± 8 μ s.

[0156] A location estimate may be referred to by other names, such as a position estimate, location, position, position fix, fix, or the like. A location estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A location estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A location estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence).

[0157] Wireless communication signals (e.g., radio frequency (RF) signals configured to carry orthogonal frequency division multiplexing (OFDM) symbols in accordance with a wireless communications standard, such as LTE, NR, etc.) transmitted between a UE and a base station can be used for environment sensing (also referred to as “RF sensing” or “radar”). Using wireless communication signals for environment sensing can be regarded as consumer-level radar with advanced detection capabilities that enable, among other things, touchless/device-free interaction with a device/system. The wireless communication signals may be cellular communication signals, such as LTE or NR signals, WLAN signals, such as Wi-Fi signals, etc. As a particular example, the wireless communication signals may be an OFDM waveform as utilized in LTE and NR. High-frequency communication signals, such as millimeter wave (mmW) RF signals, are especially beneficial to use as sensing signals because the higher frequency provides, at least, more accurate range (distance) detection.

[0158] Possible use cases of RF sensing include health monitoring use cases, such as heartbeat detection, respiration rate monitoring, and the like, gesture recognition use cases, such as human activity recognition, keystroke detection, sign language recognition, and the like, contextual information acquisition use cases, such as location detection/tracking, direction finding, range estimation, and the like, and automotive sensing use cases, such as smart cruise control, collision avoidance, and the like.

[0159] There are different types of sensing, including monostatic sensing (also referred to as “active sensing”) and bistatic sensing (also referred to as “passive sensing”). FIGS. 8A and 8B illustrate these different types of sensing. Specifically, FIG. 8A is a diagram 800 illustrating a monostatic sensing scenario and FIG. 8B is a diagram 830 illustrating a bistatic sensing scenario. In FIG. 8A, the transmitter (Tx) and receiver (Rx) are co-located in the same sensing device 804 (e.g., a UE). The sensing device 804 transmits one or more RF sensing signals 834 (e.g., uplink or sidelink positioning reference signals (PRS) where the sensing device 804 is a UE), and some of the RF sensing

signals **834** reflect off a target object **806**. The sensing device **804** can measure various properties (e.g., times of arrival (ToAs), angles of arrival (AoAs), phase shift, etc.) of the reflections **836** of the RF sensing signals **834** to determine characteristics of the target object **806** (e.g., size, shape, speed, motion state, etc.).

[0160] In FIG. **8B**, the transmitter (Tx) and receiver (Rx) are not co-located, that is, they are separate devices (e.g., a UE and a base station). Note that while FIG. **8B** illustrates using a downlink RF signal as the RF sensing signal **832**, uplink RF signals or sidelink RF signals can also be used as RF sensing signals **832**. In a downlink scenario, as shown, the transmitter is a base station and the receiver is a UE, whereas in an uplink scenario, the transmitter is a UE and the receiver is a base station.

[0161] Referring to FIG. **8B** in greater detail, the transmitter device **802** transmits RF sensing signals **832** and **834** (e.g., positioning reference signals (PRS)) to the sensing device **804**, but some of the RF sensing signals **834** reflect off a target object **806**. The sensing device **804** (also referred to as the “sensing device”) can measure the times of arrival (ToAs) of the RF sensing signals **832** received directly from the transmitter device and the ToAs of the reflections **836** of the RF sensing signals **834** reflected from the target object **806**.

[0162] More specifically, as described above, a transmitter device (e.g., a base station) may transmit a single RF signal or multiple RF signals to a sensing device (e.g., a UE). However, the receiver may receive multiple RF signals corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels.

[0163] Each path may be associated with a cluster of one or more channel taps. Generally, the time at which the receiver detects the first cluster of channel taps is considered the ToA of the RF signal on the line-of-site (LOS) path (i.e., the shortest path between the transmitter and the receiver). Later clusters of channel taps are considered to have reflected off objects between the transmitter and the receiver and therefore to have followed non-LOS (NLOS) paths between the transmitter and the receiver.

[0164] Thus, referring back to FIG. **8B**, the RF sensing signals **832** followed the LOS path between the transmitter device **802** and the sensing device **804**, and the RF sensing signals **834** followed an NLOS path between the transmitter device **802** and the sensing device **804** due to reflecting off the target object **806**. The transmitter device **802** may have transmitted multiple RF sensing signals **832**, **834**, some of which followed the LOS path and others of which followed the NLOS path. Alternatively, the transmitter device **802** may have transmitted a single RF sensing signal in a broad enough beam that a portion of the RF sensing signal followed the LOS path (RF sensing signal **832**) and a portion of the RF sensing signal followed the NLOS path (RF sensing signal **834**).

[0165] Based on the ToA of the LOS path, the ToA of the NLOS path, and the speed of light, the sensing device **804** can determine the distance to the target object(s). For example, the sensing device **804** can calculate the distance to the target object as the difference between the ToA of the LOS path and the ToA of the NLOS path multiplied by the speed of light. In addition, if the sensing device **804** is capable of receive beamforming, the sensing device **804** may be able to determine the general direction to a target

object as the direction (angle) of the receive beam on which the RF sensing signal following the NLOS path was received. That is, the sensing device **804** may determine the direction to the target object as the angle of arrival (AoA) of the RF sensing signal, which is the angle of the receive beam used to receive the RF sensing signal. The sensing device **804** may then optionally report this information to the transmitter device **802**, its serving base station, an application server associated with the core network, an external client, a third-party application, or some other sensing entity. Alternatively, the sensing device **804** may report the ToA measurements to the transmitter device **802**, or other sensing entity (e.g., if the sensing device **804** does not have the processing capability to perform the calculations itself), and the transmitter device **802** may determine the distance and, optionally, the direction to the target object **806**.

[0166] Note that if the RF sensing signals are uplink RF signals transmitted by a UE to a base station, the base station would perform object detection based on the uplink RF signals just like the UE does based on the downlink RF signals.

[0167] Like conventional radar, wireless communication-based sensing signals can be used to estimate the range (distance), velocity (Doppler), and angle (AoA) of a target object. However, the performance (e.g., resolution and maximum values of range, velocity, and angle) may depend on the design of the reference signal.

[0168] FIG. **9** illustrates an example call flow **900** for an NR-based sensing procedure (e.g., a bistatic sensing procedure) in which the network configures the sensing parameters, according to aspects of the disclosure. Although FIG. **9** illustrates a network-coordinated sensing procedure, the sensing procedure could be coordinated over sidelink channels.

[0169] At stage **905**, a sensing server **970** (e.g., inside or outside the core network) sends a request for network (NW) information to a gNB **922** (e.g., the serving gNB of a UE **904**). The request may be for a list of the UE's **904** serving cell and any neighboring cells. At stage **910**, the gNB **922** sends the requested information to the sensing server **970**. At stage **915**, the sensing server **970** sends a request for sensing capabilities to the UE **904**. At stage **920**, the UE **904** provides its sensing capabilities to the sensing server **970**. The sensing server **970** may alternatively be characterized as a sensing management function (SnMF).

[0170] At stage **925**, the sensing server **970** sends a configuration to the UE **904** indicating one or more reference signal (RS) resources that will be transmitted for sensing. The reference signal resources may be transmitted by the serving and/or neighboring cells identified at stage **910**. In some cases, the NR-based sensing procedure illustrated in FIG. **9** may be a sensing-only procedure or a joint communication and sensing (JCS) procedure. In the case of a sensing-only procedure, the reference signal resources may be reference signal resources specifically configured for sensing purposes. In the case of a JCS procedure, the reference signal resources may be reference signal resources for communication that can also be used for sensing purposes. Alternatively, the reference signal resources for sensing may be multiplexed (e.g., time-division multiplexed) with reference signal resources for communication. For example, the reference signal resources for communication may be an orthogonal frequency division multiplexing

(OFDM) waveform, while the reference signal resources for sensing may be a frequency modulation continuous wave (FMCW) waveform.

[0171] At stage 930, the sensing server 970 sends a request for sensing information to the UE 904. The UE 904 then measures the transmitted reference signals and, at stage 935, sends the measurements, or any sensing results determined from the measurements, to the sensing server 970.

[0172] In an aspect, the communication between the UE 904 and the sensing server 970 may be via the LTE positioning protocol (LPP). The communication between the sensing server 970 and the gNB may be via NR positioning protocol type A (NRPPa).

[0173] With the anticipated integration of radio frequency (RF) sensing technology in cellular systems in 5G advanced and/or 6G, positioning may leverage the RF sensing service for enhanced power savings and high localization accuracy. For example, for a user equipment (UE) which is interested in positioning, but also wants to save its power, the UE may theoretically be located using RF sensing (passive positioning) without any transmission or measurement from the UE itself. In other words, the UE itself is the sensing target. However, signaling and/or mechanisms to facilitate such passive positioning of UEs is not currently supported.

[0174] In some designs, position estimation of UEs as a sensing target may be implemented via passive UE position estimation (e.g., the target UE does not transmit or measure any signal associated with the position estimation) and semi-passive position estimation (e.g., some signals relied upon for position estimation are neither transmitted nor measured by the target UE, while other signals relied upon for the position estimation may be transmitted and/or measured by the target UE).

[0175] Aspects of the disclosure are directed to a position estimation selection scheme for position estimation of a UE. For example, depending on various factors (e.g., UE battery level, positioning accuracy requirements, etc.), different positioning schemes (e.g., PRS-only or legacy positioning, passive UE position estimation, semi-passive UE position estimation, etc.) may be preferred. In an aspect, a set of selection criteria may be evaluated to select the particular position estimation scheme to be implemented for a particular position estimation session of a UE. Such aspects may provide various technical advantages, such as reducing power consumption at the target UE (e.g., particularly advantageous for RedCap UEs), more accurate UE position estimation, reduced UE position estimation latency, and so on.

[0176] FIG. 10 illustrates an exemplary process 1000 of communications according to an aspect of the disclosure. The process 1000 of FIG. 10 is performed by a device. In some designs, the device corresponds to a UE (e.g., a target UE for position estimation, or another UE such as sidelink anchor UE or sidelink server UE). In other designs, the device corresponds to a network component (e.g., gNB/BS 304 or O-RAN component or a remote server, etc.). In scenarios where the device is integrated with another device (e.g., UE, gNB, location server, etc.), reference to any Rx/Tx operations between the respective devices may correspond to transfer of information between different logical components of the device over a data bus, etc.

[0177] Referring to FIG. 10, at 1010, the device (e.g., processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, etc.) selects a position

estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria. In an aspect, the plurality of available position estimation schemes comprising two or more of, e.g.:

[0178] a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or

[0179] a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information (e.g., DL-PRS measurement information and/or UL-PRS measurement information and/or SL-PRS measurement information, etc.) associated with one or more PRSs that are transmitted or measured by the UE, or

[0180] a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE (e.g., in an aspect, the third position estimation scheme may encompass semi-passive position estimation or joint position estimation; generally, semi-passive position estimation and joint position estimation are similar except that semi-passive position estimation relies on fewer PRS measurements, and joint position estimation assumes that sufficient PRS measurements are available for derivation of a PRS-based position estimate), or

[0181] a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate.

[0182] In some designs, a means for performing the selection of 1010 includes processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, etc., of FIGS. 3A-3C.

[0183] Referring to FIG. 10, at 1020, the device (e.g., processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, etc.) obtains location information associated with the UE in accordance with the selected position estimation scheme. In some designs, a means for obtaining the location information at 1020 includes processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, etc., of FIGS. 3A-3C.

[0184] Referring to FIG. 10, in some designs, the device further transmits a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme, and the location information is received in response to the request.

[0185] Referring to FIG. 10, in some designs, the device corresponds to a position estimation entity. In an aspect, the position estimation entity further receives measurement information associated with the selected position estimation scheme, and the location information is derived at the device based on the measurement information.

[0186] Referring to FIG. 10, in some designs, the device corresponds to the UE or another UE.

[0187] Referring to FIG. 10, in some designs, the device corresponds to a network component. In an aspect, the network component corresponds to a location management function (LMF) or a sensing management function (SnMF).

[0188] Referring to FIG. 10, in some designs, the set of selection criteria comprises, e.g.:

[0189] quality of service (QoS) information, or

[0190] load information, or

[0191] a position estimation scheme preference associated with an external device, or

[0192] position estimation scheme confidence information, or

[0193] a location region associated with the UE, or

[0194] a device type of the UE, or

[0195] a device category of the UE, or

[0196] line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or

[0197] position estimation scheme expected accuracy information, or

[0198] position estimation scheme priority information, or

[0199] any combination thereof.

[0200] Referring to FIG. 10, in some designs, the device further selects an initial position estimation scheme that is different from the selected position estimation scheme, and the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0201] Referring to FIG. 10, in some designs, the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme. In an aspect, the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme. In an aspect, the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0202] FIG. 11 illustrates an exemplary process 1100 of communications according to an aspect of the disclosure. The process 1100 of FIG. 11 is performed by a device. In some designs, the device corresponds to a UE (e.g., a target UE for position estimation, or another UE such as sidelink anchor UE or sidelink server UE). In other designs, the device corresponds to a network component (e.g., gNB/BS 304 or O-RAN component or a remote server, etc.). In scenarios where the device is integrated with another device (e.g., UE, gNB, location server, etc.), reference to any Rx/Tx operations between the respective devices may correspond to transfer of information between different logical components of the device over a data bus, etc.

[0203] Referring to FIG. 11, at 1110, the device (e.g., receiver 312 or 322 or 352 or 362, network transceiver(s) 380 or 390, etc.) receives an indication of a selected position estimation scheme for position estimation of a user equipment (UE). In an aspect, the selected position estimation scheme corresponding to one of:

[0204] a first position estimation scheme that provides a first position estimate based on first radio frequency for

sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or

[0205] a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information (e.g., DL-PRS measurement information and/or UL-PRS measurement information and/or SL-PRS measurement information, etc.) associated with one or more PRSs that are transmitted or measured by the UE, or

[0206] a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE (e.g., in an aspect, the third position estimation scheme may encompass semi-passive position estimation or joint position estimation; generally, semi-passive position estimation and joint position estimation are similar except that semi-passive position estimation relies on fewer PRS measurements, and joint position estimation assumes that sufficient PRS measurements are available for derivation of a PRS-based position estimate), or

[0207] a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate.

[0208] In some designs, a means for performing the reception of 1110 includes processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, etc., of FIGS. 3A-3C.

[0209] Referring to FIG. 11, at 1120, the device (e.g., processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, receiver 312 or 322 or 352 or 362, transmitter 314 or 324 or 354 or 364, etc.) performs a position estimation session of the UE in accordance with the selected position estimation scheme. In an aspect, the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme. In an aspect, the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme. In an aspect, the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme. In some designs, a means for obtaining the location information at 1120 includes processor 342 or 384 or 394, sensing and/or positioning component 348 or 388 or 398, receiver 312 or 322 or 352 or 362, transmitter 314 or 324 or 354 or 364, etc., of FIGS. 3A-3C.

[0210] Referring to FIG. 11, in some designs, the device corresponds to a position estimation entity.

[0211] Referring to FIG. 11, in some designs, the device corresponds to the UE.

[0212] Referring to FIG. 11, in some designs, the device corresponds to the LMF.

[0213] Referring to FIG. 11, in some designs, the device corresponds to the SnMF.

[0214] Referring to FIG. 11, in some designs, the device further receives an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme, and the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0215] Referring to FIG. 11, in some designs, the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme. In an aspect, the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0216] Referring to FIG. 11, in some designs, the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0217] Referring to FIGS. 10-11, in a specific example as noted above, the difference between active and semi-passive positioning is that for semi-passive positioning, the measurements and/or transmissions that the UE performs may not be sufficient for good positioning accuracy. However, by combining with RF sensing operations, accurate UE location information can be derived. In a further aspect, in joint passive and active positioning, both entities (LMF and SnMF) perform UE positioning, and the outcome is fused to generate the final UE location information. In an aspect, this mode can occur either by, e.g.:

[0218] Option 1 [output-level fusion]: fusing location estimate derived independently at both entities.
Example: $x_{UE} = \alpha \hat{x}_{LMF} + (1 - \alpha) \hat{x}_{SnMF}$, for $\alpha \in [0, 1]$

[0219] Option 2 [measurements-level fusion]: The sensing measurements, and the positioning measurements are collected at one entity (e.g., SnMF) and the final UE location information is derived.

[0220] Referring to FIGS. 10-11, in a specific example, the sensing management function (SnMF) may be implemented as an independent core network entity, separate from the LMF given the different services provided, and given the virtualization and the software-defined networking (SDN) philosophies in the 5G/6G network core design. Therefore, in an aspect, any signaling between these network entities may be standardized for proper scalability and core network optimal operations.

[0221] Referring to FIGS. 10-11, in a specific example, a QoS-based selection may be implemented. For example, depending on the quality of service (QoS) requirement of the UE location information request received by the LMF, a subset of the modes are allowed to be used by the LMF. Below, uncertainty definitions are provided for further context.

[0222] In some designs, a method of describing the uncertainty for latitude and longitude has been sought which is both flexible (can cover wide differences in range) and efficient. The proposed solution makes use of a variation on the Binomial expansion. The uncertainty r , expressed in metres, is mapped to a number K , with the following formula:

$$r = C((1 + x)^K - 1) \quad \text{Equation 1}$$

with $C=10$ and $x=0, 1$. With $0 \leq K \leq 127$, a suitably useful range between 0 and 1800 kilometres is achieved for the uncertainty, while still being able to code down to values as small as 1 metre. The uncertainty can then be coded on 7 bits, as the binary encoding of K .

[0223] Example values for K are as follows, e.g.:

TABLE 1

Example values for the uncertainty Function	
Value of K	Value of uncertainty
0	0 m
1	1 m
2	2.1 m
—	—
20	57.3 m
—	—
40	443 m
—	—
60	3 km
—	—
80	20 km
—	—
100	138 km
—	—
120	927 km
—	—
127	1800 km

[0224] For high accuracy uncertainty r , Equation 1 may also be utilized, with $C=0.3$ and $x=0.02$. With $0 \leq K \leq 255$, a suitably useful range between 0 and 46.49129 metres is achieved for the high accuracy uncertainty, while still being able to code down to values as small as 6 millimetre. The uncertainty can then be coded on 8 bits, as the binary encoding of K , e.g.:

TABLE 2

Example values for the high accuracy uncertainty function	
Value of K	Value of uncertainty
0	0 m
1	0.006 m
2	0.01212 m
—	—
20	0.14578 m
—	—
40	0.36241 m
—	—
60	0.68430 m
—	—
80	1.16263 m
—	—
100	1.87339 m
—	—
120	2.92954 m
—	—
127	3.40973 m
—	—
255	46.49129 m

[0225] In this example, depending on the ‘High Accuracy Uncertainty’ level K in the QoS, the following may be inferred, e.g.:

TABLE 3

K vs. Positioning Mode	Joint	Active	Semi-Passive	Passive
$K \leq 60$	YES	YES	NO	NO
$60 < K \leq 120$	YES	YES	YES	NO
$K > 120$	YES	YES	YES	YES

[0226] Referring to FIGS. 10-11, in a specific example, a load-based selection may be implemented. For example, depending on the “load” of the LMF, the LMF may choose to operate in one of the positioning modes. The load of the LMF can be defined using multiple ways. One possible definition is the number of ongoing positioning sessions N_s , the LMF is serving. Depending on N_s , the LMF may choose a positioning mode for an incoming UE location information request.

[0227] Referring to FIGS. 10-11, in a specific example, the LMF may select the positioning mode based on a UE-initiated or a 3rd party request. In an aspect, the UE-initiated request can be motivated by power saving, while maintaining the positioning service. Note that the LMF may not fulfill the request based on the request (e.g., different parameters may be selected or the request may be denied).

[0228] Referring to FIGS. 10-11, in a specific example, when operating in a passive or semi-passive positioning mode, the SnMF may not be able to confidently locate the target and indicate that to the LMF. Based on the SnMF indication, the LMF can switch from passive or semi-passive positioning, into legacy active positioning mode or joint mode. There may be various reasons that may prevent the SnMF from accurately localizing the UE including the presence of multiple nearby targets creating ambiguities in the passive sensing, unavailability of sensing nodes in the area around the target UE, etc.

[0229] Referring to FIGS. 10-11, in a specific example, in some areas, it may be easier to locate an object passively. For example, a connected car driving in a highway. In this case, the RF sensing reliability is higher and the LMF can rely on passive/semi-passive positioning without sacrificing positioning QoS. In an aspect, the selection of the positioning depends on the target UE location. For example, there can areas for which passive/semi-positioning may be activated by the LMF. In a specific example, a connected car is located using active positioning within the city, and when the car leaves the city and is on the highway, the LMF switches to passive/semi-passive positioning to save the UE power.

[0230] Referring to FIGS. 10-11, in a specific example, passive/semi-passive positioning may be selected only for certain types or categories of UEs (e.g., UEs which can be sensed efficiently, e.g., a connected car).

[0231] Referring to FIGS. 10-11, in a specific example, the LMF selection of the positioning mode can be a function of any combination of the triggers described above (e.g., including the QoS and the LMF load, the UE request, infeasibility or non-reliability of passive positioning, and the UE location, etc.).

[0232] Referring to FIGS. 10-11, in a specific example, the LMF can dynamically activate/switch between positioning mode of operation of the UE. For example, depending on the scenario and the trigger conditions as highlighted in previous slides, the LMF may decide to switch between active, semi-passive, and passive positioning. In an aspect, the switching indication can be signaled to the UE through LPP.

In an aspect, the activation of one mode can be temporary. In an aspect, the LMF may configure a UE with a new positioning mode and a positioning mode time window, such that the termination of the time window, the UE autonomously switches back to the default mode. In an aspect, the LMF may configure the active positioning as the default positioning mode. In an aspect, a time window can be indicated in the activation configuration from the LMF, or in the semi-passive/passive positioning mode assistance data.

[0233] Referring to FIGS. 10-11, in a specific example for UE-based semi-passive positioning, the UE can be the center point for calculating the location coordinates in which it requests a few active positioning (e.g., using PRS/SRS signals) from LMF and sends RF sensing request directly to SnMF to obtain passive sensing results. The UE can then leverage the measured active positioning outcomes and fuse them with the ones provided by the SnMF. For example, the UE may have a better algorithm/implementation to fuse active and passive positioning outcomes, and/or SnMF and LMF may be from different vendors, such that the UE may have better collaboration with them and hence can be well-vested to obtain better positioning outcome for semi-passive positioning.

[0234] Referring to FIGS. 10-11, in a specific example for prioritization/adaptation of UE-based passive/semi-passive/active/joint positioning, the LMF can also provide some recommendations to help UE decide on when to prioritize/adapt between the UE-based active, passive, and semi-passive positioning. Examples of such assistance may include (1) LOS/NLOS conditions of UE with respect to anchors (TRPs or UEs) to be used for the active positioning, (2) Expected accuracy for different positioning modes, (3) Flag indication on recommended positioning modes, and (4) ordered list of recommended positioning modes. In an aspect, the above assistance options can be associated with an area ID.

[0235] In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorporated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an electrical insulator and an electrical conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

[0236] Implementation examples are described in the following numbered clauses:

[0237] Clause 1. A method of operating a device, comprising: selecting a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtaining location information associated with the UE in accordance with the selected position estimation scheme.

[0238] Clause 2. The method of clause 1, further comprising: transmitting a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme, wherein the location information is received in response to the request.

[0239] Clause 3. The method of any of clauses 1 to 2, wherein the device corresponds to a position estimation entity, further comprising: receiving measurement information associated with the selected position estimation scheme, wherein the location information is derived at the device based on the measurement information.

[0240] Clause 4. The method of any of clauses 1 to 3, wherein the device corresponds to the UE or another UE, or the device corresponds to a network component, wherein the network component corresponds to a location management function (LMF), or wherein the network component corresponds to a sensing management function (SnMF).

[0241] Clause 5. The method of any of clauses 1 to 4, wherein the set of selection criteria comprises: quality of service (QoS) information, or load information, or a position estimation scheme preference associated with an external device, or position estimation scheme confidence information, or a location region associated with the UE, or a device type of the UE, or a device category of the UE, or line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or position estimation scheme expected accuracy information, or position estimation scheme priority information, or any combination thereof.

[0242] Clause 6. The method of any of clauses 1 to 5, further comprising: selecting an initial position estimation scheme that is different from the selected position estimation scheme, wherein the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0243] Clause 7. The method of any of clauses 1 to 6, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0244] Clause 8. The method of clause 7, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0245] Clause 9. The method of any of clauses 7 to 8, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0246] Clause 10. A method of operating a device, comprising: receiving an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and performing a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0247] Clause 11. The method of clause 10, wherein the device corresponds to a position estimation entity, or wherein the device corresponds to the UE, or wherein the device corresponds to the LMF, or wherein the device corresponds to the SnMF, or a combination thereof.

[0248] Clause 12. The method of any of clauses 10 to 11, further comprising: receiving an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme, wherein the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0249] Clause 13. The method of any of clauses 10 to 12, wherein the selected position estimation scheme is associ-

ated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0250] Clause 14. The method of any of clauses 10 to 13, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0251] Clause 15. The method of any of clauses 10 to 14, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0252] Clause 16. A device, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: select a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtain location information associated with the UE in accordance with the selected position estimation scheme.

[0253] Clause 17. The device of clause 16, wherein the one or more processors, either alone or in combination, are further configured to: transmit, via the one or more transceivers, a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme, wherein the location information is received in response to the request.

[0254] Clause 18. The device of any of clauses 16 to 17, wherein the device corresponds to a position estimation entity, further comprising: receive, via the one or more transceivers, measurement information associated with the selected position estimation scheme, wherein the location information is derived at the device based on the measurement information.

[0255] Clause 19. The device of any of clauses 16 to 18, wherein the device corresponds to the UE or another UE, or the device corresponds to a network component, wherein the network component corresponds to a location management function (LMF), or wherein the network component corresponds to a sensing management function (SnMF).

[0256] Clause 20. The device of any of clauses 16 to 19, wherein the set of selection criteria comprises: quality of service (QoS) information, or load information, or a posi-

tion estimation scheme preference associated with an external device, or position estimation scheme confidence information, or a location region associated with the UE, or a device type of the UE, or a device category of the UE, or line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or position estimation scheme expected accuracy information, or position estimation scheme priority information, or any combination thereof.

[0257] Clause 21. The device of any of clauses 16 to 20, wherein the one or more processors, either alone or in combination, are further configured to: select an initial position estimation scheme that is different from the selected position estimation scheme, wherein the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0258] Clause 22. The device of any of clauses 16 to 21, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0259] Clause 23. The device of clause 22, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0260] Clause 24. The device of any of clauses 22 to 23, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0261] Clause 25. A device, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: receive, via the one or more transceivers, an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and perform a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected posi-

tion estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0262] Clause 26. The device of clause 25, wherein the device corresponds to a position estimation entity, or wherein the device corresponds to the UE, or wherein the device corresponds to the LMF, or wherein the device corresponds to the SnMF, or a combination thereof.

[0263] Clause 27. The device of any of clauses 25 to 26, wherein the one or more processors, either alone or in combination, are further configured to: receive, via the one or more transceivers, an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme, wherein the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0264] Clause 28. The device of any of clauses 25 to 27, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0265] Clause 29. The device of any of clauses 25 to 28, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0266] Clause 30. The device of any of clauses 25 to 29, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0267] Clause 31. A device, comprising: means for selecting a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and means for obtaining location information associated with the UE in accordance with the selected position estimation scheme.

[0268] Clause 32. The device of clause 31, further comprising: means for transmitting a request to a position estimation entity for the position estimation of the UE in

accordance with the selected position estimation scheme, wherein the location information is received in response to the request.

[0269] Clause 33. The device of any of clauses 31 to 32, wherein the device corresponds to a position estimation entity, further comprising: means for receiving measurement information associated with the selected position estimation scheme, wherein the location information is derived at the device based on the measurement information.

[0270] Clause 34. The device of any of clauses 31 to 33, wherein the device corresponds to the UE or another UE, or the device corresponds to a network component, wherein the network component corresponds to a location management function (LMF), or wherein the network component corresponds to a sensing management function (SnMF).

[0271] Clause 35. The device of any of clauses 31 to 34, wherein the set of selection criteria comprises: quality of service (QoS) information, or load information, or a position estimation scheme preference associated with an external device, or position estimation scheme confidence information, or a location region associated with the UE, or a device type of the UE, or a device category of the UE, or line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or means for positioning estimation scheme expected accuracy information, or means for positioning estimation scheme priority information, or any combination thereof.

[0272] Clause 36. The device of any of clauses 31 to 35, further comprising: means for selecting an initial position estimation scheme that is different from the selected position estimation scheme, wherein the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0273] Clause 37. The device of any of clauses 31 to 36, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0274] Clause 38. The device of clause 37, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0275] Clause 39. The device of any of clauses 37 to 38, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0276] Clause 40. A device, comprising: means for receiving an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement

information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and means for performing a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0277] Clause 41. The device of clause 40, wherein the device corresponds to a position estimation entity, or wherein the device corresponds to the UE, or wherein the device corresponds to the LMF, or wherein the device corresponds to the SnMF, or a combination thereof.

[0278] Clause 42. The device of any of clauses 40 to 41, further comprising: means for receiving an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme, wherein the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0279] Clause 43. The device of any of clauses 40 to 42, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0280] Clause 44. The device of any of clauses 40 to 43, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0281] Clause 45. The device of any of clauses 40 to 44, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0282] Clause 46. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a device, cause the device to: select a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third

position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and obtain location information associated with the UE in accordance with the selected position estimation scheme.

[0283] Clause 47. The non-transitory computer-readable medium of clause 46, further comprising computer-executable instructions that, when executed by the device, cause the device to: transmit a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme, wherein the location information is received in response to the request.

[0284] Clause 48. The non-transitory computer-readable medium of any of clauses 46 to 47, wherein the device corresponds to a position estimation entity, further comprising: receive measurement information associated with the selected position estimation scheme, wherein the location information is derived at the device based on the measurement information.

[0285] Clause 49. The non-transitory computer-readable medium of any of clauses 46 to 48, wherein the device corresponds to the UE or another UE, the device corresponds to a network component, or wherein the network component corresponds to a location management function (LMF), or wherein the network component corresponds to a sensing management function (SnMF).

[0286] Clause 50. The non-transitory computer-readable medium of any of clauses 46 to 49, wherein the set of selection criteria comprises: quality of service (QoS) information, or load information, or a position estimation scheme preference associated with an external device, or position estimation scheme confidence information, or a location region associated with the UE, or a device type of the UE, or a device category of the UE, or line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or position estimation scheme expected accuracy information, or position estimation scheme priority information, or any combination thereof.

[0287] Clause 51. The non-transitory computer-readable medium of any of clauses 46 to 50, further comprising computer-executable instructions that, when executed by the device, cause the device to: select an initial position estimation scheme that is different from the selected position estimation scheme, wherein the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0288] Clause 52. The non-transitory computer-readable medium of any of clauses 46 to 51, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0289] Clause 53. The non-transitory computer-readable medium of clause 52, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0290] Clause 54. The non-transitory computer-readable medium of any of clauses 52 to 53, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0291] Clause 55. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a device, cause the device to: receive an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of: a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and perform a position estimation session of the UE in accordance with the selected position estimation scheme, wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation scheme, the third position estimation scheme or the fourth position estimation scheme.

[0292] Clause 56. The non-transitory computer-readable medium of clause 55, wherein the device corresponds to a position estimation entity, or wherein the device corresponds to the UE, or wherein the device corresponds to the LMF, or wherein the device corresponds to the SnMF, or a combination thereof.

[0293] Clause 57. The non-transitory computer-readable medium of any of clauses 55 to 56, further comprising computer-executable instructions that, when executed by the device, cause the device to: receive an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme, wherein the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

[0294] Clause 58. The non-transitory computer-readable medium of any of clauses 55 to 57, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

[0295] Clause 59. The non-transitory computer-readable medium of any of clauses 55 to 58, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

[0296] Clause 60. The non-transitory computer-readable medium of any of clauses 55 to 59, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

[0297] Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0298] Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0299] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0300] The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside

in an ASIC. The ASIC may reside in a user terminal (e.g., UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

[0301] In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0302] While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. For example, the functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Further, no component, function, action, or instruction described or claimed herein should be construed as critical or essential unless explicitly described as such. Furthermore, as used herein, the terms “set,” “group,” and the like are intended to include one or more of the stated elements. Also, as used herein, the terms “has,” “have,” “having,” “comprises,” “comprising,” “includes,” “including,” and the like does not preclude the presence of one or more additional elements (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”) or the alternatives are mutually exclusive (e.g., “one or more” should not be interpreted as “one and more”). Furthermore, although components, functions, actions, and instructions may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Accordingly, as used herein, the articles “a,” “an,” “the,” and “said” are intended to include one or

more of the stated elements. Additionally, as used herein, the terms “at least one” and “one or more” encompass “one” component, function, action, or instruction performing or capable of performing a described or claimed functionality and also “two or more” components, functions, actions, or instructions performing or capable of performing a described or claimed functionality in combination.

What is claimed is:

1. A method of operating a device, comprising:
 - selecting a position estimation scheme for a position estimation session of a user equipment (UE) from a plurality of available position estimation schemes based on a set of selection criteria, the plurality of available position estimation schemes comprising two or more of:
 - a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or
 - a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or
 - a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or
 - a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and
 - obtaining location information associated with the UE in accordance with the selected position estimation scheme.
2. The method of claim 1, further comprising:
 - transmitting a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme, wherein the location information is received in response to the request.
3. The method of claim 1,
 - wherein the device corresponds to a position estimation entity, further comprising:
 - receiving measurement information associated with the selected position estimation scheme, wherein the location information is derived at the device based on the measurement information.
4. The method of claim 1,
 - wherein the device corresponds to the UE or another UE, or
 - the device corresponds to a network component, wherein the network component corresponds to a location management function (LMF), or
 - wherein the network component corresponds to a sensing management function (SnMF).
5. The method of claim 1, wherein the set of selection criteria comprises:

quality of service (QOS) information, or
 load information, or
 a position estimation scheme preference associated with
 an external device, or
 position estimation scheme confidence information, or
 a location region associated with the UE, or
 a device type of the UE, or
 a device category of the UE, or
 line of sight (LOS) or non-LOS (NLOS) information
 associated with the UE and one or more wireless nodes,
 or
 position estimation scheme expected accuracy informa-
 tion, or
 position estimation scheme priority information, or
 any combination thereof.

6. The method of claim 1, further comprising:

selecting an initial position estimation scheme that is
 different from the selected position estimation scheme,
 wherein the selection of the selected position estimation
 scheme transitions the position estimation session from
 the initial position estimation scheme to the selected
 position estimation scheme.

7. The method of claim 1, wherein the selected position
 estimation scheme is associated with at least one triggering
 event that triggers a transition from the selected position
 estimation scheme to another position estimation scheme.

8. The method of claim 7, wherein the at least one
 triggering event comprises termination of a position estima-
 tion scheme time window or signaling that requests a
 transition to the another position estimation scheme.

9. The method of claim 7, wherein the at least one
 triggering event is signaled via long term evolution posi-
 tioning protocol (LPP), or configuration information, or
 assistance data, or any combination thereof.

10. A method of operating a device, comprising:

receiving an indication of a selected position estimation
 scheme for position estimation of a user equipment
 (UE), the selected position estimation scheme corre-
 sponding to one of:

a first position estimation scheme that provides a first
 position estimate based on first radio frequency for
 sensing (RF-S) measurement information associated
 with one or more RF-S signals that are transmitted
 and measured by one or more wireless nodes other
 than the UE, or

a second position estimation scheme that provides a
 second position estimate based only on positioning
 reference signal (PRS) measurement information
 associated with one or more PRSs that are transmit-
 ted or measured by the UE, or

a third position estimation scheme that provides a third
 position estimate based on a combination of two or
 more of the first RF-S measurement information, the
 PRS measurement information, and second RF-S
 measurement information associated with at least
 one RF-S signal that is transmitted, measured, or
 both, by the UE, or

a fourth position estimation scheme that provides a
 fourth position estimate that is based on the first
 position estimate and the second position estimate;
 and

performing a position estimation session of the UE in
 accordance with the selected position estimation
 scheme,

wherein the device corresponds to the UE and the selected
 position estimation scheme corresponds to the first
 position estimation scheme, the third position estima-
 tion scheme or the fourth position estimation scheme,
 or

wherein the device corresponds to a location management
 function (LMF) and the selected position estimation
 scheme corresponds to the first position estimation
 scheme, the third position estimation scheme or the
 fourth position estimation scheme, or

wherein the device corresponds to a sensing management
 function (SnMF) and the selected position estimation
 scheme corresponds to the second position estimation
 scheme, the third position estimation scheme or the
 fourth position estimation scheme.

11. The method of claim 10,

wherein the device corresponds to a position estimation
 entity, or

wherein the device corresponds to the UE, or

wherein the device corresponds to the LMF, or

wherein the device corresponds to the SnMF.

12. The method of claim 10, further comprising:

receiving an initial indication of an initial position esti-
 mation scheme that is different from the selected posi-
 tion estimation scheme,

wherein the indication of the selected position estimation
 scheme transitions the position estimation session from
 the initial position estimation scheme to the selected
 position estimation scheme.

13. The method of claim 10, wherein the selected position
 estimation scheme is associated with at least one triggering
 event that another position estimation scheme triggers a
 transition from the selected position estimation scheme to
 another position estimation scheme.

14. The method of claim 13, wherein the at least one
 triggering event comprises termination of a position estima-
 tion scheme time window or signaling that requests a
 transition to the another position estimation scheme.

15. The method of claim 10, wherein the at least one
 triggering event is signaled via long term evolution posi-
 tioning protocol (LPP), or configuration information, or
 assistance data, or any combination thereof.

16. A device, comprising:

one or more memories;

one or more transceivers; and

one or more processors communicatively coupled to the
 one or more memories and the one or more transceiv-
 ers, the one or more processors, either alone or in
 combination, configured to:

select a position estimation scheme for a position estima-
 tion session of a user equipment (UE) from a plurality
 of available position estimation schemes based on a set
 of selection criteria, the plurality of available position
 estimation schemes comprising two or more of:

a first position estimation scheme that provides a first
 position estimate based on first radio frequency for
 sensing (RF-S) measurement information associated
 with one or more RF-S signals that are transmitted and
 measured by one or more wireless nodes other than the
 UE, or

a second position estimation scheme that provides a
 second position estimate based only on positioning

reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or

a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or

a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and

obtain location information associated with the UE in accordance with the selected position estimation scheme.

17. The device of claim 16, wherein the one or more processors, either alone or in combination, are further configured to:

transmit, via the one or more transceivers, a request to a position estimation entity for the position estimation of the UE in accordance with the selected position estimation scheme,

wherein the location information is received in response to the request.

18. The device of claim 16,

wherein the device corresponds to a position estimation entity, further comprising:

receive, via the one or more transceivers, measurement information associated with the selected position estimation scheme,

wherein the location information is derived at the device based on the measurement information.

19. The device of claim 16,

wherein the device corresponds to the UE or another UE, or

the device corresponds to a network component,

wherein the network component corresponds to a location management function (LMF), or

wherein the network component corresponds to a sensing management function (SnMF).

20. The device of claim 16, wherein the set of selection criteria comprises:

quality of service (QOS) information, or

load information, or

a position estimation scheme preference associated with an external device, or

position estimation scheme confidence information, or

a location region associated with the UE, or

a device type of the UE, or

a device category of the UE, or

line of sight (LOS) or non-LOS (NLOS) information associated with the UE and one or more wireless nodes, or

position estimation scheme expected accuracy information, or

position estimation scheme priority information, or

any combination thereof.

21. The device of claim 16, wherein the one or more processors, either alone or in combination, are further configured to:

select an initial position estimation scheme that is different from the selected position estimation scheme,

wherein the selection of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

22. The device of claim 16, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

23. The device of claim 22, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

24. The device of claim 22, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

25. A device, comprising:

one or more memories;

one or more transceivers; and

one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to:

receive, via the one or more transceivers, an indication of a selected position estimation scheme for position estimation of a user equipment (UE), the selected position estimation scheme corresponding to one of:

a first position estimation scheme that provides a first position estimate based on first radio frequency for sensing (RF-S) measurement information associated with one or more RF-S signals that are transmitted and measured by one or more wireless nodes other than the UE, or

a second position estimation scheme that provides a second position estimate based only on positioning reference signal (PRS) measurement information associated with one or more PRSs that are transmitted or measured by the UE, or

a third position estimation scheme that provides a third position estimate based on a combination of two or more of the first RF-S measurement information, the PRS measurement information, and second RF-S measurement information associated with at least one RF-S signal that is transmitted, measured, or both, by the UE, or

a fourth position estimation scheme that provides a fourth position estimate that is based on the first position estimate and the second position estimate; and

perform a position estimation session of the UE in accordance with the selected position estimation scheme,

wherein the device corresponds to the UE and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or

wherein the device corresponds to a location management function (LMF) and the selected position estimation scheme corresponds to the first position estimation scheme, the third position estimation scheme or the fourth position estimation scheme, or

wherein the device corresponds to a sensing management function (SnMF) and the selected position estimation scheme corresponds to the second position estimation

scheme, the third position estimation scheme or the fourth position estimation scheme.

26. The device of claim **25**,

wherein the device corresponds to a position estimation entity, or

wherein the device corresponds to the UE, or

wherein the device corresponds to the LMF, or

wherein the device corresponds to the SnMF.

27. The device of claim **25**, wherein the one or more processors, either alone or in combination, are further configured to:

receive, via the one or more transceivers, an initial indication of an initial position estimation scheme that is different from the selected position estimation scheme,

wherein the indication of the selected position estimation scheme transitions the position estimation session from the initial position estimation scheme to the selected position estimation scheme.

28. The device of claim **25**, wherein the selected position estimation scheme is associated with at least one triggering event that triggers a transition from the selected position estimation scheme to another position estimation scheme.

29. The device of claim **28**, wherein the at least one triggering event comprises termination of a position estimation scheme time window or signaling that requests a transition to the another position estimation scheme.

30. The device of claim **25**, wherein the at least one triggering event is signaled via long term evolution positioning protocol (LPP), or configuration information, or assistance data, or any combination thereof.

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